



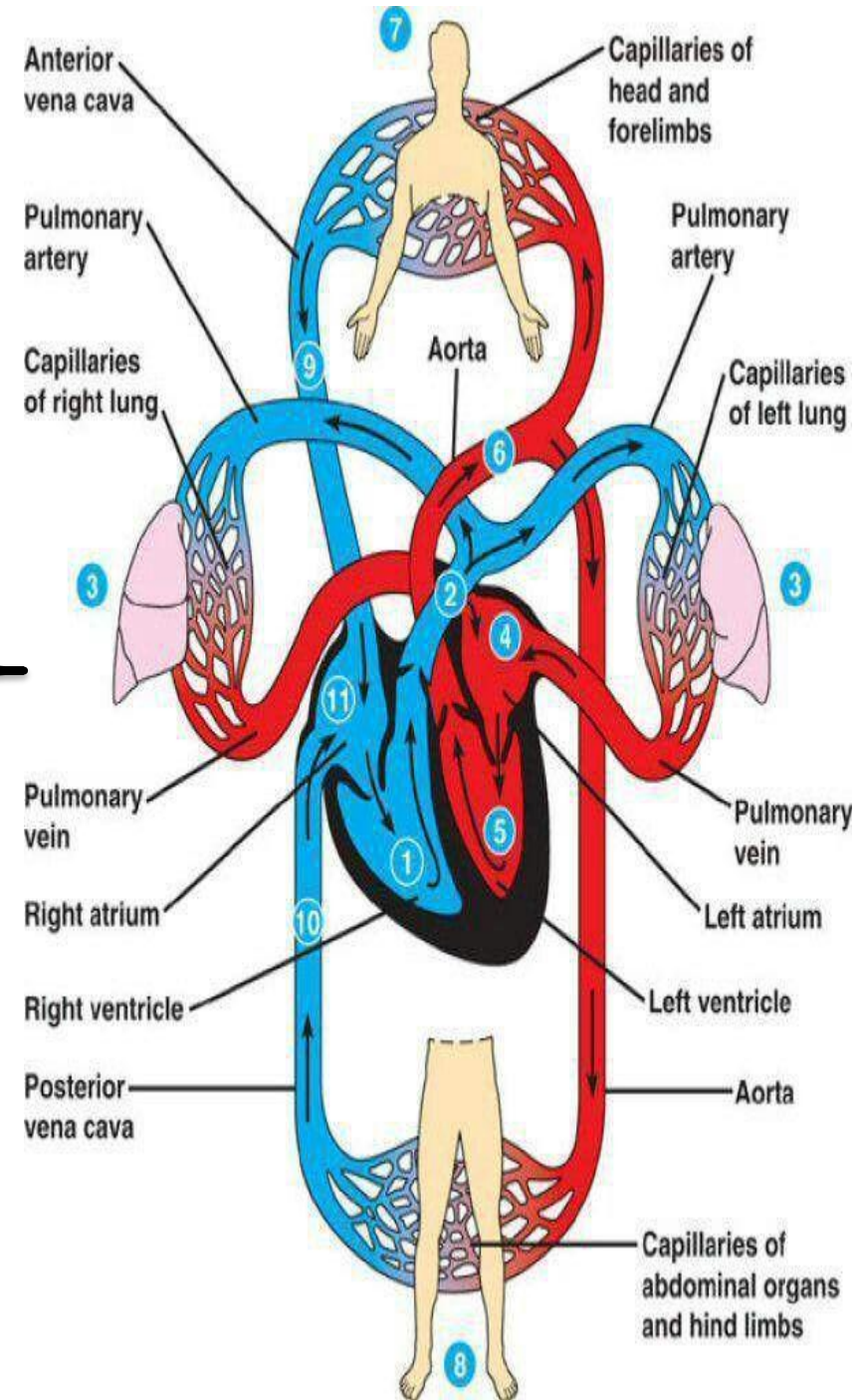
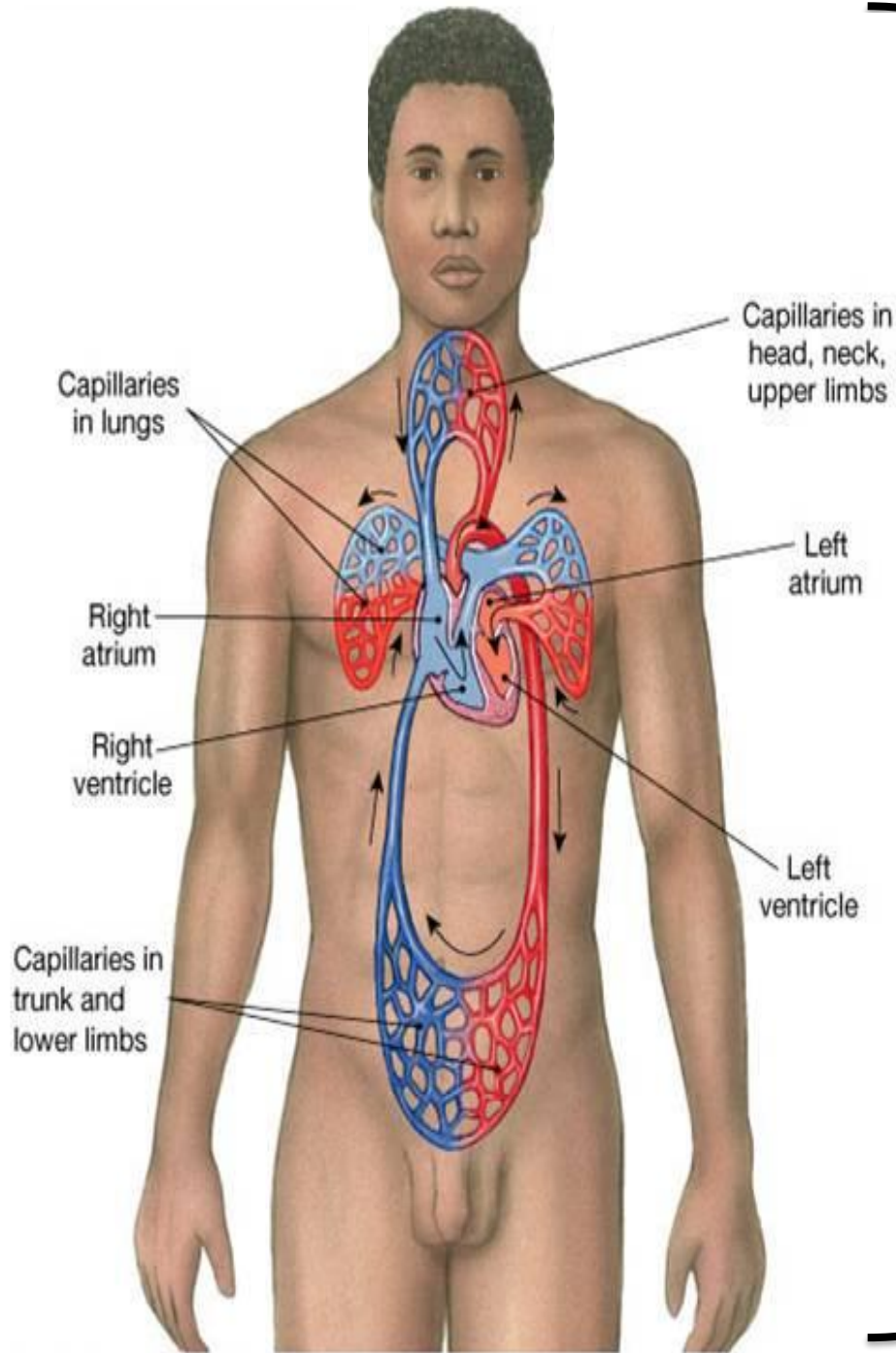
SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY !



Physiology of Cardio-Vascular System [CVS]

Fikremariam. A [Medical Physiology]

[2016 E.C]



Learning objectives

At the end of this chapter, students should be able to:

- ☐ Recognize some terminologies [~ CVS]
- ☐ Cardiovascular system [CVS]↓
 - ☐ Functions
 - ☐ Components of the CVS

1. Heart (Def.)

- ☐ Characteristics of heart cells [myocardial cells]
- ☐ Layers of heart wall
- ✓ Pericardium → Myocardium → Endocardium
 - ✓ Chambers [4]
 - ✓ Valves [leaflet] [4]
- ☐ Conduction system [conduction pathways]
 - ✓ Contraction [S] Vs relaxation [D]
 - ✓ Ejecting [S] Vs filling [D]
- ☐ Cardiac cycle [CC]
 - ✓ Stroke volume [SV]
 - ✓ Heart rate [HR]
 - ✓ Cardiac output [CO]
 - ✓ Heart sounds (HSS)
 - ✓ Cardiac reserve (CR):
- ☐ ECG /EKG

☐ Circulation

- ✓ Types of circulation [2]

2. Blood vessels

- ☐ Types [3]
- ☐ Functions
- ☐ Blood supply to the heart itself [5%]
 - ✓ By coronary artery
- ☐ Blood pressure [BP]
 - ✓ Measurement [2]
 - ✓ Factors affecting BP
- ☐ Hypertension [HT]
 - ✓ Stages [grades] of HT
 - ✓ Etiology [Types] of HT
 - ✓ Risk factors for HT
- ☐ Regulation of BP [2]
- ☐ Treatment for HT
- ☐ Angina Pectoris????
- ☐ Heart Failure????

Terminologies

1. Membrane potential (E_m)
2. RMP
 - ✓ Polarization
3. Depolarization
4. Repolarization
5. **Threshold potential** → lower limit Or minimum potential to reach AP
6. **Action potential = AP** = (nerve impulse, nerve signal) = wave of depolarization = depolarization current
7. SA node
8. AV-node
9. Syncytium → Networked
1. ANP → Atrial Natriuretic Peptide
2. ADH → Antidiuretic Hormone [Vasopressin]
3. Ischemia (MI = Myocardial infarction) → Heart failure
4. Systole Vs diastole
5. Auscultation
6. Sphygmomanometer
7. Stethoscope
8. ECF volume
9. Artery → Capillary → Vein
10. Sympathetic Vs parasympathetic [PS]
11. Stenosis
12. Regurgitation

Cardiovascular System



- ❑ A system of pumping organ and Vessels that transport blood to & from tissues throughout the body.

Functions



1. Transport & distribute essential materials (subs) to all parts of the body
 - ✓ Bringing & supply O₂ and nutrients
2. Remove metabolic by-products (wastes).
3. Regulation of body temperature.
4. Hormonal function [secret ANP]



Transport in the CVS

Materials entering the body	Moved from	To
O ₂	Lungs	All cells
Nutrients & water	Intestinal tract	All cells
Materials leaving the body	From	To
Metabolic wastes	All cells	Kidneys
Heat	All cells	Skin
CO ₂	All cells	Lungs
Materials moved from cell to cell	From cell	To cell
Immune cells [Abs]	Blood cells	Available to any cells that needs them
Hormones	Endocrine cells	Target cells
Stored nutrients [GLU]	Liver cells	All cells

Components of CVS



1. **Heart:** Central pumping organ

2. **Blood vessels:**

- ✓ **Arteries** →→→ Distributing system (Oxygenated blood)
- ✓ **Capillaries** →→→ Exchange system
- ✓ **Veins** →→→ Collecting system (Deoxygenated blood)

NB:Exceptions → Pulmonary artery & Umbilical artery (Deoxy) Vs Pulmonary veins & Umbilical veins (Oxy).

Heart



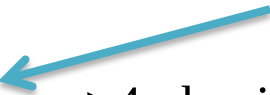
- ❑ Weighs about 300g in adults (about the size of a clenched fist)
 - ✓ Located in the **Mediastinum**
- ❑ Is hollow, muscular organ that pumps blood to all parts of the body through 2 ‘Circulations’.
 1. **Pulmonary (Lungs) &**
 2. **Systemic** (the rest of the body except pulmonary circulation).

Heart Cells [Myocardial Cells]



- ❑ Are striated (consists of sarcomeres just like skeletal muscle)
- ❑ Have numerous mitochondria (up to 40% of cell volume) → always on work.
- ❑ Adjacent cells join end-to-end by **intercalated discs**
 - ❑ **Intercalated discs** contain 2 types of specialized junctions:
 - ✓ Desmosomes [allow fluids to pass]
 - ✓ Gap junctions [allow small sized subs and ions [AP]

Characteristics of Myocardial Cells



- ❑ Have >4 physiological properties, which are essential for the function of the heart.

1. Automaticity (Autorhythmicity):

- ✓ Is the ability of the heart to generate (initiate) cardiac impulse and to beat regularly independent of any external stimulus.
- ✓ Without involving the nervous system

2. Excitability : Ability to respond to an electrical stimulation/to an impulse [AP]

3. Conductivity : passes or propagates electrical impulses from cell to cell.

- ✓ Ability to transmit an impulse.

4. Contractility : shortens in response to electrical stimulation

~~✓ Ability to respond with pumping action (relax Vs contract)~~

5. Extensibility : The ability to be stretched (extended)

6. Elasticity : The ability to return to original shape after contraction or extension

Anatomy & Histology of the Heart

Heart is made up of 3 distinct wall layers [Layers of Heart Wall]

1 . Endocardium

- ✓ Innermost layer
- ✓ Thin serous membrane, composed of unique type of epithelial tissue, that lines the interior of the heart.

2 . Myocardium [Cardiac Muscle]

- ✓ The middle thickest **Muscular layer** of the heart wall
 - ✓ i.e. the bulk of the heart wall.

3. Pericardium

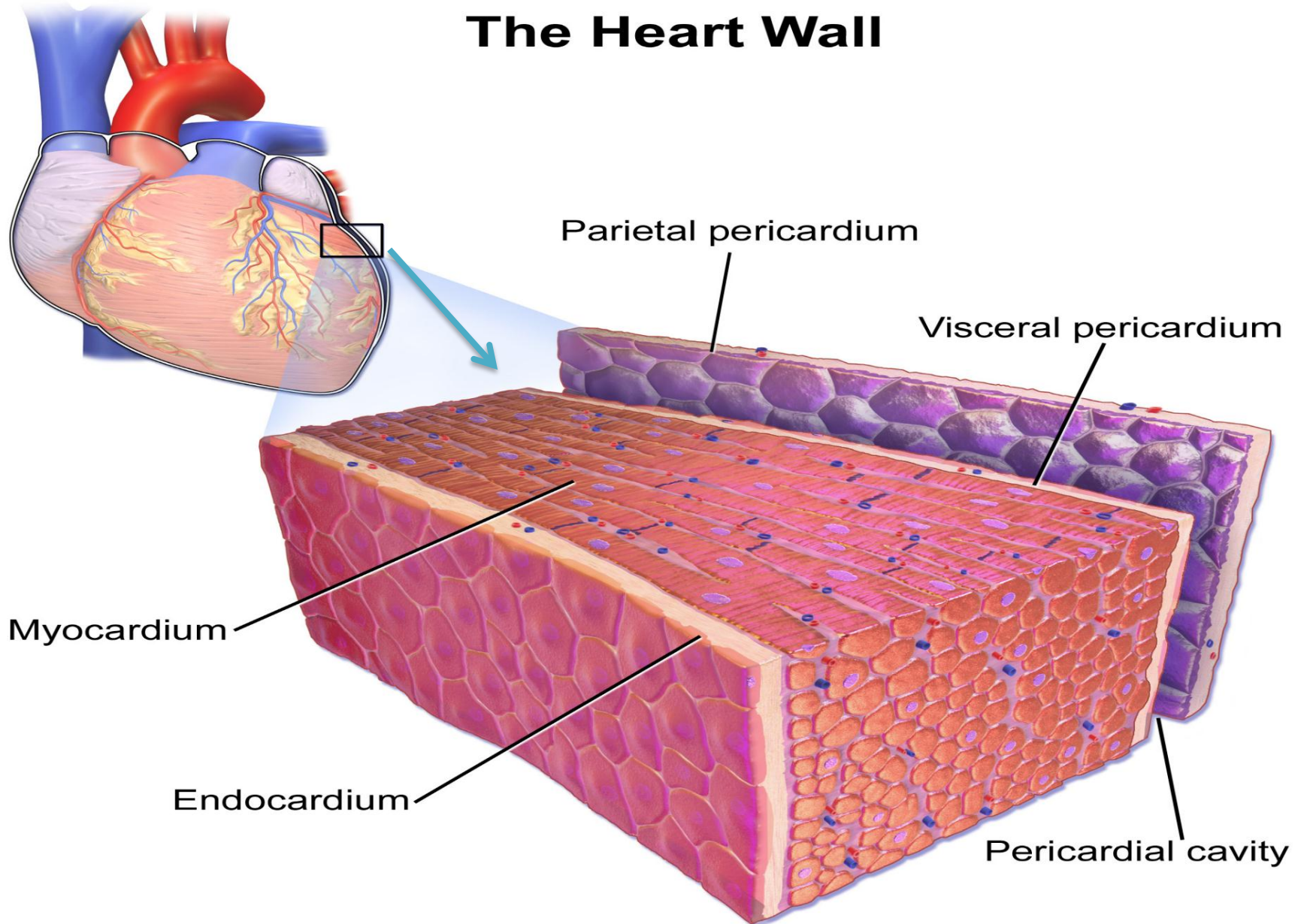
3.1 . Visceral Pericardium [Epicardium]

- ✓ The innermost layers of the pericardium [has actual contact with the surface of the heart Or it closely envelops the heart]
- ✓ Thin, external membrane around the heart
 - ✓ Allow protection against friction (rubbing with organs).

3.2.Parietal Pericardium: The outermost layers of the pericardium

Cont'd

The Heart Wall



Heart chambers:

❑ The human heart consists of 4 chambers:

- ✓ 2 thin walled **atria** (right & left) form the upper part of the heart &
- ✓ 2 thick walled **ventricles** (right & left) form the lower part.

❑ The 2 atria are separated by an **inter-atrial septum** and

❑ The 2 ventricles by an **inter-ventricular septum**.

❑ Each of the 4 chambers of the heart contract and relax rhythmically.

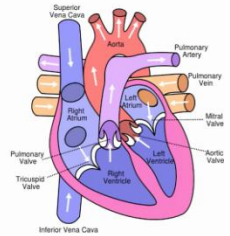
✓ Filling with blood during **Diastole**.

✓ Ejecting the blood during **Systole**.

❑ The right and left heart **contract and relax** simultaneously.

✓ Ejecting equal blood volume at the same time but,

✓ With different pressure.



GoAnimate

Syncytium (as a unit)

❑ Cardiac muscle tissue forms 2 functional syncytia or units

1. The **atria** being one &
2. The **ventricles** the other.

❑ Bcz of the presence of gap junctions, if any cell is stimulated within a syncytium, the impulse will spread to all cells.

- ✓ The 2 atria always function as a unit &
- ✓ The 2 ventricles always function as a unit.

❑ However, there are no gap junctions b/n atrial & ventricular contractile cells.

- ✓ The atria & ventricles are separated by the electrically nonconductive tissue
 - ✓ Fibrous tissue or septum → (weak or slow conductor)
 - ✓ It is the only electrical window b/n Atria & ventricles
 - ✓ We call it **AV nodal** delay [silence]

Heart Valves

- ❑ Prevent back flow of blood and ensure that blood flows in the proper unidirection.
- ❑ Valves open & close in response to changes in pressure (pressure gradient).

1. Atrioventricular (AV) valves ➡ prevent backflow of blood from ventricles to atria during ventricular systole (contraction).

❑ AV valves

- ✓ Open when pressure [**P**] in the atria is greater than 'P' in the ventricles (i.e., during ventricular diastole) &
- ✓ Closed when 'P' in the ventricles is greater than 'P' in the atria (i.e., during ventricular systole) ,Includes

1. Tricuspid valve - located b/n Rt atrium & Rt ventricle

- ✓ Prevent back flow from Rt. ventricle to Rt. atrium during ventricular systole.

2. Bicuspid (mitral valve) - located b/n Lt atrium & Lt ventricle

- ✓ Prevent back flow from Lt .Ventricle to Lt. atrium during ventricular systole)

Cont'd

2.Semilunar (SL) valves

- ❑ Prevent backflow of blood from arteries (Pulmonary artery & the Aorta) to ventricles during ventricular diastole (relaxation).
- ✓ Open when 'P' in the ventricles is greater than 'P' in the arteries (i.e., during ventricular systole) and
- ✓ Closed when 'P' in the pulmonary trunk & aorta is greater than 'P' in the ventricles (i.e. during ventricular diastole).

They are

1. **Aortic Valve** ➡ b/n Lt ventricle & the aorta
2. **Pulmonary Valve** ➡ b/n Rt ventricle & the pulmonary artery (trunk)

Cardiac Muscle (Myocardium)

❑ In cardiac muscle there are 2 types of cells:

(based on functional criteria)

1. Autorhythmic (Automatic) cells

- ✓ Are self-stimulating cells without any external stimulation
- ✓ Have specialized excitatory and conductive systems having 2 functions
 - ✓ **Generate** excitatory impulse and **send** to working myocardial cells
- 1. The specialized **excitatory** cells → → → They generate rhythmic cardiac impulse
 - ✓ Are pacemakers
- 2. **Conductive** muscle fibres → → → They conduct cardiac impulse
 - ✓ Are conductive systems

2. Contractile cells (Atrial & Ventricular muscle)

- ✓ Working myocardium (atria & ventricles)
 - ✓ Do mechanical work of pumping
- ✓ Contract when stimulated.

Cont'd

Cardiac cells	Functions
1. Autorhythmic cells ✓ Excitatory cells ✓ Conductive cells	✓ Generate impulse (Pacemaker) ✓ Conduct impulse
2. Contractile cells Atria and Ventricle	✓ For contraction (pump) when stimulated ✓ Working myocardium ✓ Do mechanical work of pumping

Conductive System

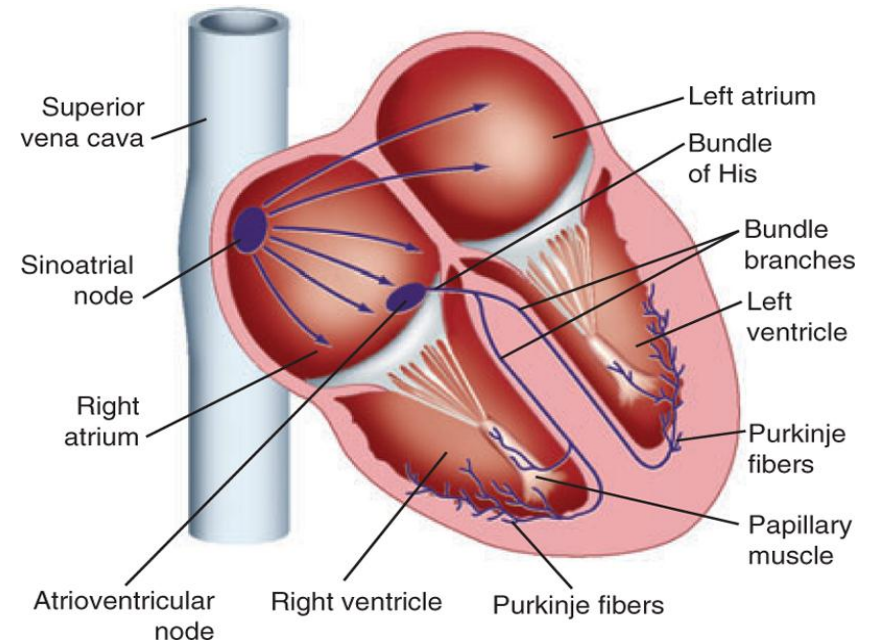
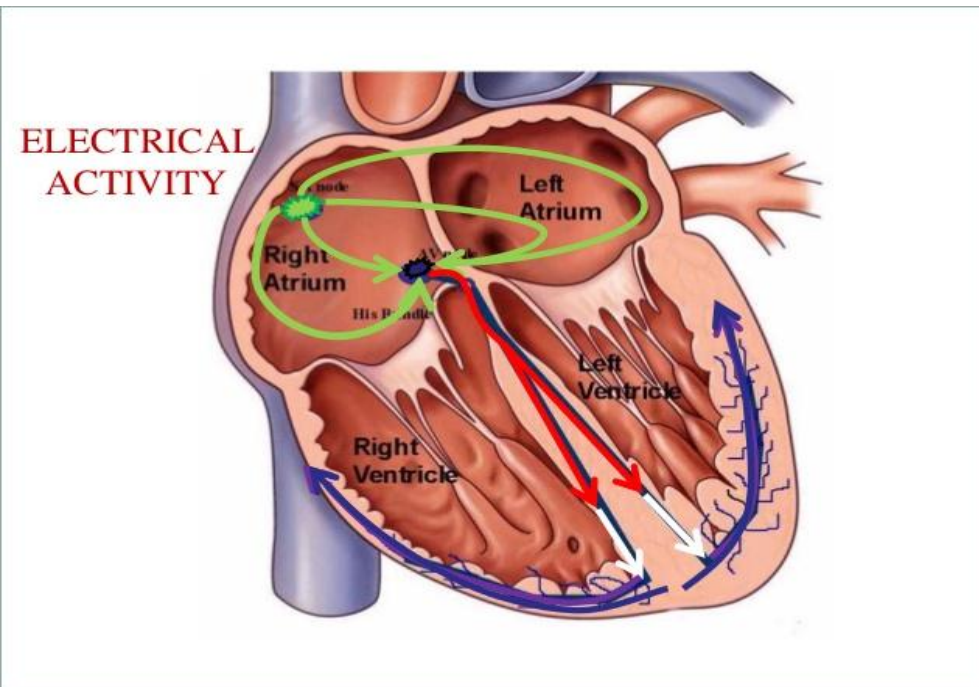
Origin and Spread of Excitation

❑ Atria and ventricle behave functionally as a **syncytium**

✓ Excitation anywhere in atria or ventricles spread all over unexcited fibers.

✓ **Intercalated discs** : offer no obstacle to conduction of excitation

The cardiac conduction system.



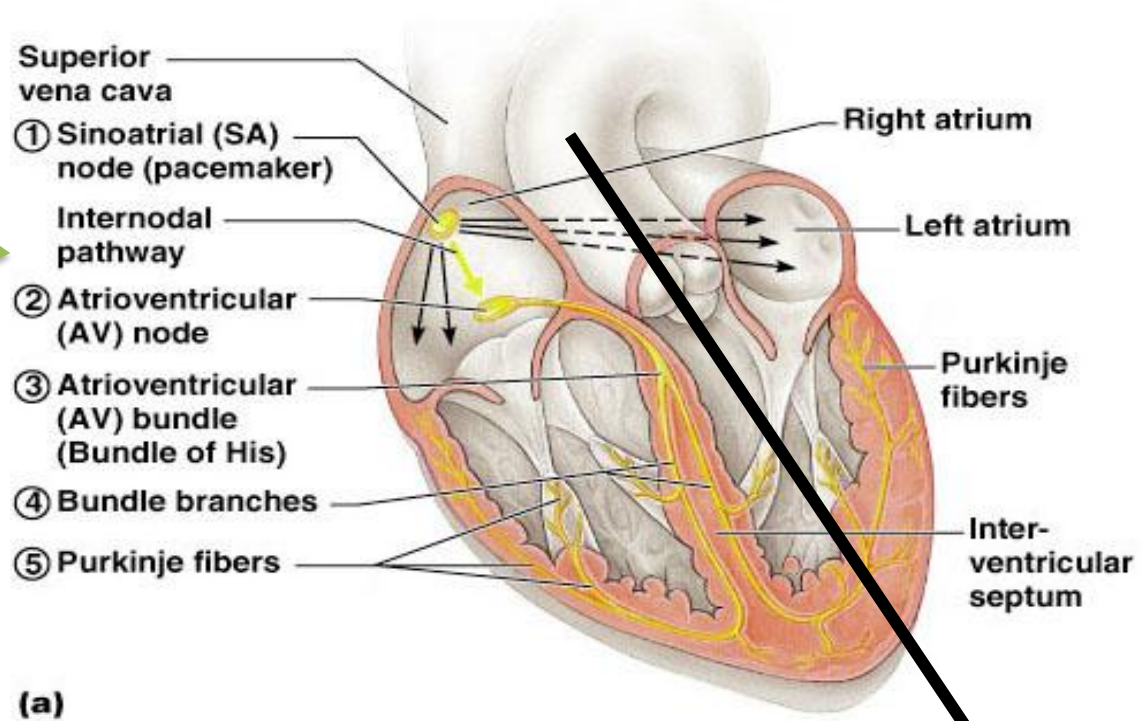
Specialized excitatory and conductive system of the heart

1. **SA-node:** [The fastest ,the primary and dominant cardiac pacemaker]
 - ✓ Is small mass of specialized cardiac muscle located in the posterior wall of the RA.
 - ✓ It acts as a normal pacemaker by generating the electric impulses of the heartbeat → Drive heart at a rate of 72bpm (at rest) → at regular intervals
 - ✓ Is where a normal rhythmical impulse is generated
 - ✓ Excitation from **SA-node** spreads over working myocardium of atria → ventricles
 - ✓ Intrinsic rhythmicity of **SA node** is greater than other pacemaker sites
 - ✓ Firing of **SA node** suppresses automaticity in other loci
2. **Internodal pathways:** conduct impulse from the SA-node to the AV-node
3. **AV-node:** [secondary pacemaker] =The only pathway for conduction to ventricles.
 - ✓ Is a site where impulse from the atria delayed to be conducted to the ventricle
 - ✓ Propagation delayed at AV node (site of nodal delay)
 - ✓ Is potential pacemaker (if SA node fails)
4. **AV bundle (bundle of His):** which conducts impulse from the atria [AV node] to ventricle.
5. **Purkinje fibers:** They conduct cardiac impulse to the ventricles: Rhythm=20-40 rhythm/min

Summary [Geometry of Propagation ➔ Sequence of Excitation]

The Authorhythmic cells

1. Sinoatrial (SA), or sinus node
2. Internodal pathways
3. AV node
4. AV bundle (bundle of His)
5. Right & left bundle branches
6. Purkinje fibers



Various Automatic cells have different rhythms

Normal rate of AP discharge in Automatic tissue of the heart

Tissue

AP [impulse/minute]

1. SA node (normal pace maker)
2. AV node
3. Bundle of His (branches) & purkinje fibers

80-120 bpm (usually 70-80)
 40-60 rhythm/min or bpm or AP
 20-40

Cardiac Cycle

- ❑ Is the cardiac event that occurs from the beginning of 1 heart beat to the beginning of the next beat.
- ❑ Is sequence of events (activities) in a single beat
- ❑ Is repeating pattern of contraction-relaxation of cardiac chambers in the heart
 - ✓ Where the 2 atria (RA & LA) contract and shortly after the 2 ventricles (RV & LV) contract simultaneously → It consists of period of contraction called “systole” & period of relaxation “diastole”.
- ❑ **A single cardiac cycle** comprised of =systole + diastole
- ❑ Assume a heart rate of **72bpm**, the average duration of 1 cardiac cycle is =0.8 sec (60/72) duration.
- ❑ Systolic (contraction) =0.3 sec & diastolic (relaxation)=0.5 sec duration.

What activities accomplished in a cardiac cycle?

- ✓ Heart sounds, ECG tracing, Atrial & Ventricular pressure
- ✓ VV, VEDV, VESV

$$1\text{CC} = 1\text{beat} = \text{systole} + \text{diastole} = 1\text{PQRST}$$

Important Terms



❑ Ventricular Volumes (VV): The volume of blood in the ventricles

❑ Ventricular End Diastolic Volume (VEDV)

✓ The volume of blood in the ventricle at the end of ventricular diastole (relaxation phase)

✓ EDV = 120-140ml [Average ~ 120]

❑ Ventricular End Systolic Volume (VESV):

✓ The volume of blood that remains in the ventricle at the end of ventricular systole (contraction phase)

✓ ESV = 50-60 ml [Average ~ 50]

❑ Heart Rate (HR)

✓ Is the number of cardiac cycles (heart beats) per minute

✓ Is the rate at which the heart beats.

✓ Normal HR: 60 -100 beats/minute (Average ~72bpm)

✓ Low HR [< 60 bpm] is Called **Bradycardia** but normal in athletes

✓ High HR [> 100 bpm] is Called **Tachycardia**

Cont'd

□ Stroke Volume (SV)

- ✓ Is the volume of blood pumped by each ventricle per **beat**.
- ✓ Is the difference of **end diastolic volume and end systolic volume**
- ✓ $SV = EDV - ESV \Rightarrow 120 - 50 = 70 \text{ ml/beat}$ (Average $\sim 70\text{-}80 \text{ ml/beat}$)
 - ✓ EDV is volume of blood in the ventricle at the end of diastole [Filling] and
 - ✓ ESV is the volume at the end of systole [ejection]

□ Cardiac output

- ✓ The volume of blood ejected (pumped) from the heart (the ventricles) per **minute**.

$$CO = SV \times HR = 5.5 \text{ L/min}$$

□ SV = is expressed in **ml/beat** and

□ HR = in **beats/minute** → Therefore

□ CO is in **ml/minute**.

- ✓ Usually about 5,500 ml (or 5.5 liters) per minute

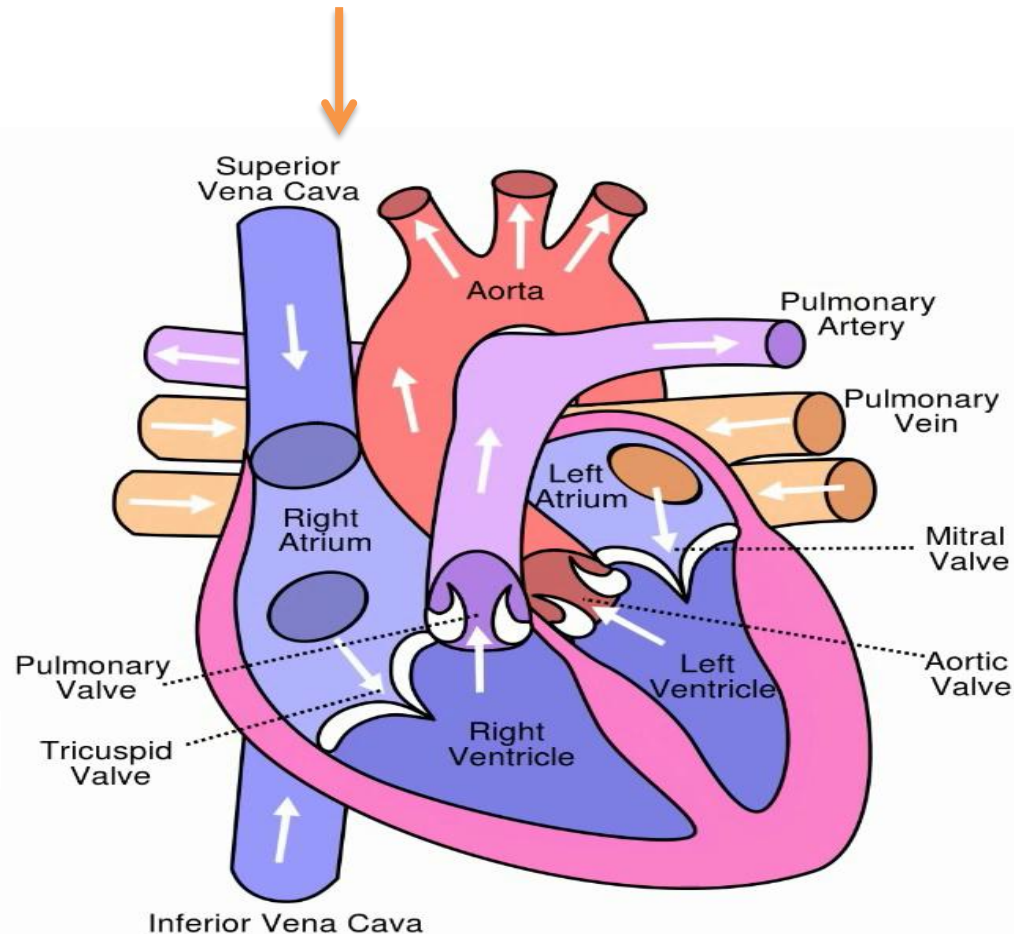
Cont'd

Cardiac Reserve (CR):

- ❑ Is the %age difference b/n the maximal exercise CO and the resting CO
- ❑ It is the maximum percentage that the CO can increase above normal
 - ✓ CO = 5-6 L/minute at rest
 - ✓ CO = 20-40 L/minute at maximum exercise
- ❑ Thus, in healthy young adult, the CR is about 300 to 400%.
- ❑ CR permits CO to increase radically during periods of physical activity.

Heart sounds

- ❑ Sounds that are created by rapid **closure of valves** and vibrations.
- ✓ There are 4 valves which can be closed and opened in response to pressure changes during each cardiac cycle.



Cont'd

1. Atrioventricular (AV) valves

- ❑ Prevent backflow of blood from ventricles to atria during ventricular systole (contraction).
 - ✓ **Tricuspid valve** - located b/n Rt atrium & Rt ventricle.
 - ✓ **Mitral valve** - located b/n Lt atrium & Lt ventricle.
- ❑ Open when 'P' in the atria is greater than 'P' in the ventricles
 - ✓ (i.e., during ventricular diastole).
- ❑ Closed when 'P' in the ventricles is greater than 'P' in the atria
 - ✓ (i.e., during ventricular systole).
 - ✓ Thus, the 1st heart sound is produced during this period.

Cont'd

2.Semilunar [SL] valves

❑ Prevent backflow of blood from arteries OR

✓ From pulmonary artery & the Aorta to ventricles during ventricular diastole (relaxation).

✓ **Aortic valve** - located b/n LV & the aorta

✓ **Pulmonary valve** - located b/n RV & the P.Artery (Trunk).

❑ Open when 'P' in the ventricles is greater than 'P' in the arteries

✓ (i.e. during ventricular systole).

❑ Closed when 'P' in the pulmonary trunk & aorta is greater than 'P' in the ventricles

✓ (i.e. during ventricular diastole).

✓ Thus ,the 2nd heart sound is produced during this period.

First Heart Sound (S₁)

❑ Is produced due to sudden closure of AV valves.

- ✓ Tricuspid and mitral valves.
- ✓ It Occurs at the beginning of systole.
- ✓ Sounds like LUB.
- ✓ Soft, lower pitched (25-45Hz) and long in duration(0.14s) as compared to S₂.

Best site of Auscultation

- ✓ The tricuspid component is heard at the 4th intercostal space **left** to sternum.
- ✓ The mitral component is heard **at the apex** (**left** 5th intercostal space mid clavicular line)

Second Heart Sound (S₂)

- ❑ Is produced by Sudden closure of
 - ✓ SL valves [Aortic and Pulmonic].
 - ✓ Occurs at the beginning of diastole.
 - ✓ Sounds like DUB.
 - ✓ Sharper, louder (50Hz), and shorter in duration (0.11 second).

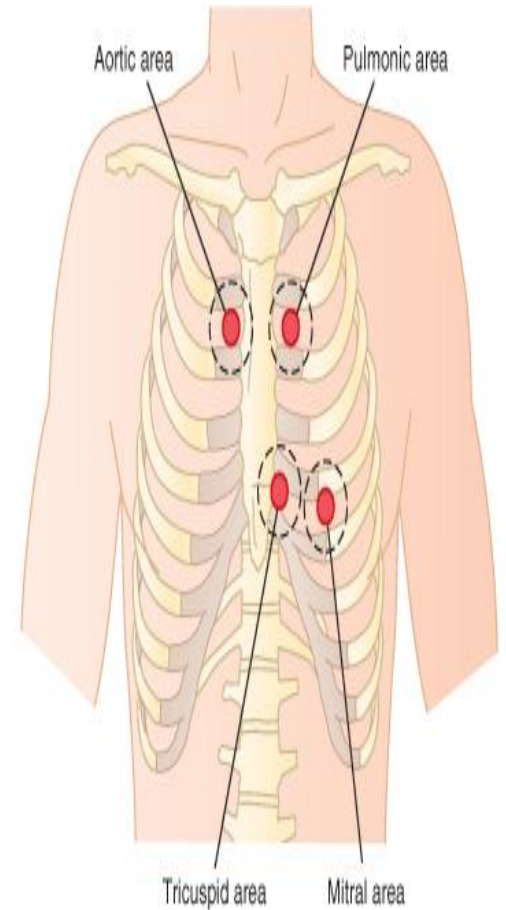
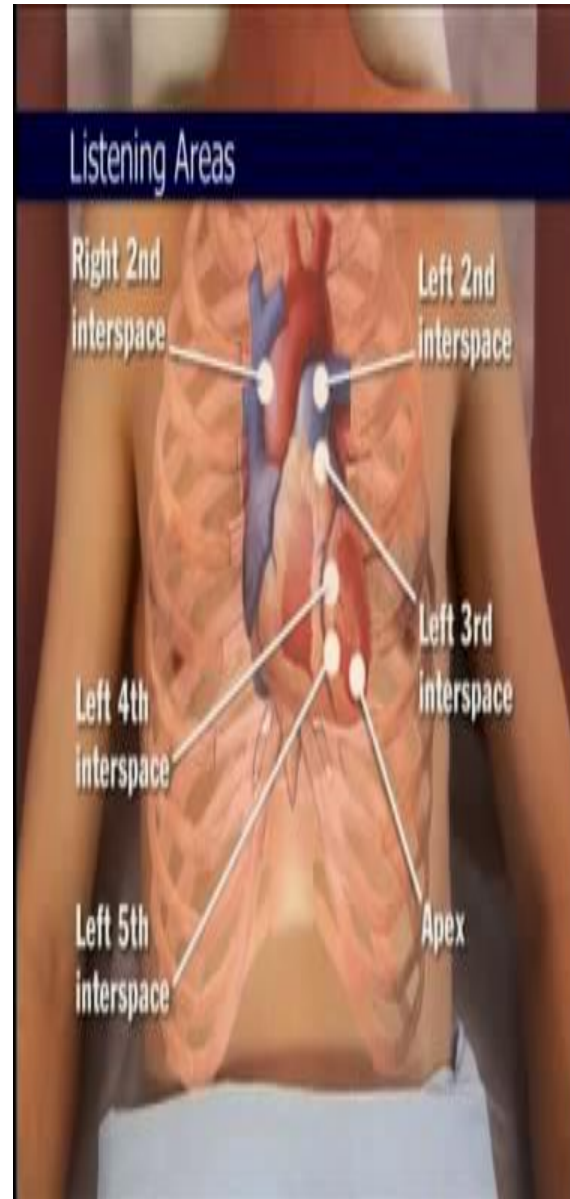
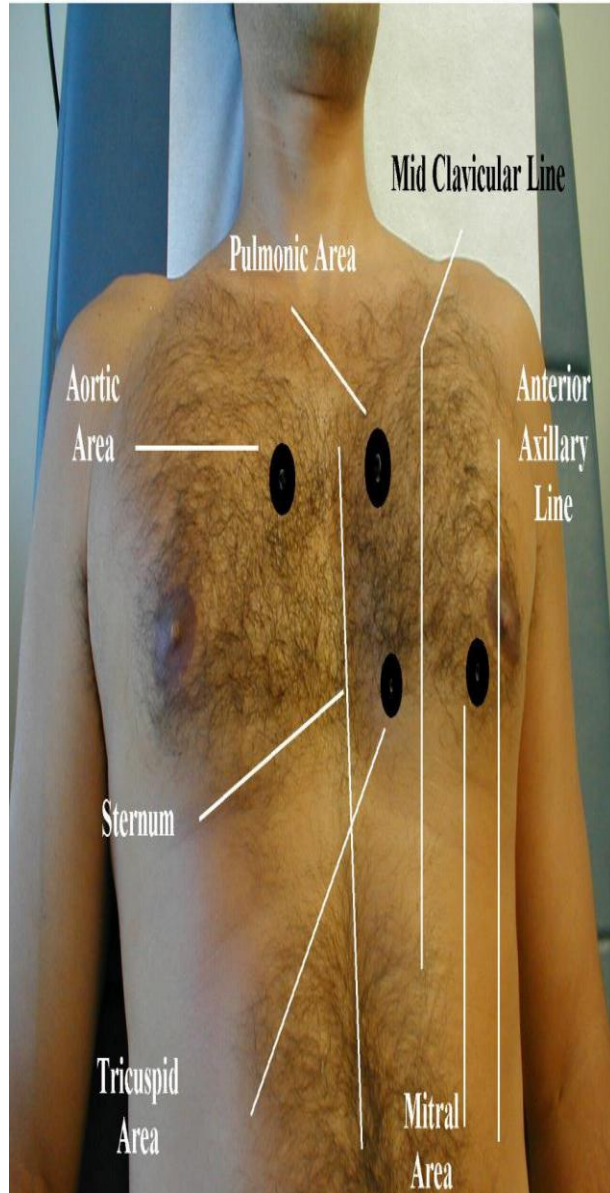
Best site of Auscultation

- ✓ The pulmonary component is heard at the **2nd left intercostal** space.
- ✓ The aortic component is heard at the **2nd right intercostal** space.

- ❑ **S1 and S2** Auscultated in 4 valvular areas with **diaphragm**.

- ✓ Are distinct sounds heard (Have high pitched sound).

Areas of Auscultation



er. Guyton & Hall: Textbook of Medical Physiology 11e - www.studentconsult.com

Third heart sound (S₃)

- ❑ Is produced due to the vibration that created by the **rapid flowing** of blood from atria to ventricles (during ventricular diastole)
 - ❑ Caused by blood from LA colliding with "left over" blood in LV
 - ❑ Occurs at the beginning of diastole (**rapid ventricular filling**).
 - ❑ Low pitched sound which is difficult to hear with stethoscope.
- **S₃ follows S₂**
 - ✓ It is audible in **children** and in adults during **exercise**
 - ✓ Best site of auscultation is ➔ at the **apex** of the heart beneath the Xiphoid process.

Fourth heart sound (S₄)

❑ Produced due to atrial contraction which causes rapid flow of blood to ventricles and vibration in the blood.

✓ When blood pressed (squeezed) into non-compliant LV.

✓ Occurs at the end of ventricular diastole

✓ Low pitched

✓ S₄ precedes S₁

↓
Cause: rapid ventricular filling during atrial systole

Timing: during atrial systole

Best site of auscultation: at the apex of the heart

Summary [Heart Sounds]

- ❑ Heart sounds (lubb-dubb) are associated with **closure** of heart valves
- ❑ 4-separate audible heart sounds (S_1 , S_2 , S_3 and S_4)

Means of identification

1. **Auscultation**: direct stethoscope mediated listening
2. **Phonocardiographic** based recording

S_1 : Is always audible & ➡➡ has a LUBB-sound

S_2 : Is always audible & ➡➡ has DUBB-sound

✓ S_1 & S_2 [Continuous heart sound]: Lubb-Dubb, Lubb-Dubb , Lubb-Dubb.....

✓ Duration of Dubb [S_2] is **shorter & louder** than that of Lubb

S_3 : Is audible in **children** and in adults during **exercise**

S_4 : Is audible very rarely

Murmurs

❑ Are abnormal heart sounds produced (created)

- ✓ By excessive degree of turbulence BF in the heart chambers or across valves.
- ✓ When there is an abnormality of heart valves (stenosis and regurgitation)

❑ Classified as

❑ Systolic murmur (Left heart)

- ✓ Mitral regurgitation
- ✓ Aortic stenosis

Diastolic murmur (Left heart)

- Mitral stenosis
- Aortic regurgitation

Division of the Circulation

❑ In the CVS, blood passes [flow=transport] through 2 (double) circulations:

1. **Systemic circulation**

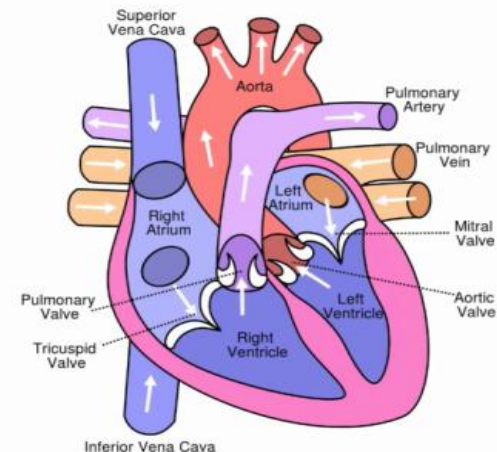
2. **Pulmonary circulation**

1. Systemic circulation:

❑ The general circulation of the blood from the heart to all body parts (except the lungs) and back to the heart.

❑ Circulation that supplies blood to all the body except to the lungs

❑ It starts from the LV → Aorta → Systemic arteries → Systemic capillaries → Veins → SVC & IVC → ends in RA



SVC=Superior vena cava

IVC=inferior vena cava

2. Pulmonary circulation:

- ❑ The passage of blood from the right side of the heart through arteries to the lungs.
 - ✓ It picks up oxygen and is returned to the left side of the heart by veins
- ❑ Starts in the RV → Pulmonary trunk → Pulmonary arteries → Pulmonary capillaries → Pulmonary veins → ends in the LA.

Circulation

A process by which nutrients, respiratory gases, and metabolic products are transported throughout the body.

Summary [Systemic & Pulmonary circulation]



- ❑ Blood returning from the **systemic (body) circulation** enters the right atrium (via the inferior & superior vena cave).



- ❑ From there, blood flows into the right ventricle, which then pumps blood to



- ❑ The lungs (via the **pulmonary arteries**=**the only arteries that carry deoxygenated blood in adult**)



- ❑ Blood returning from the lungs enters the left atrium (via 4 pulmonary veins) then to



The left ventricle.

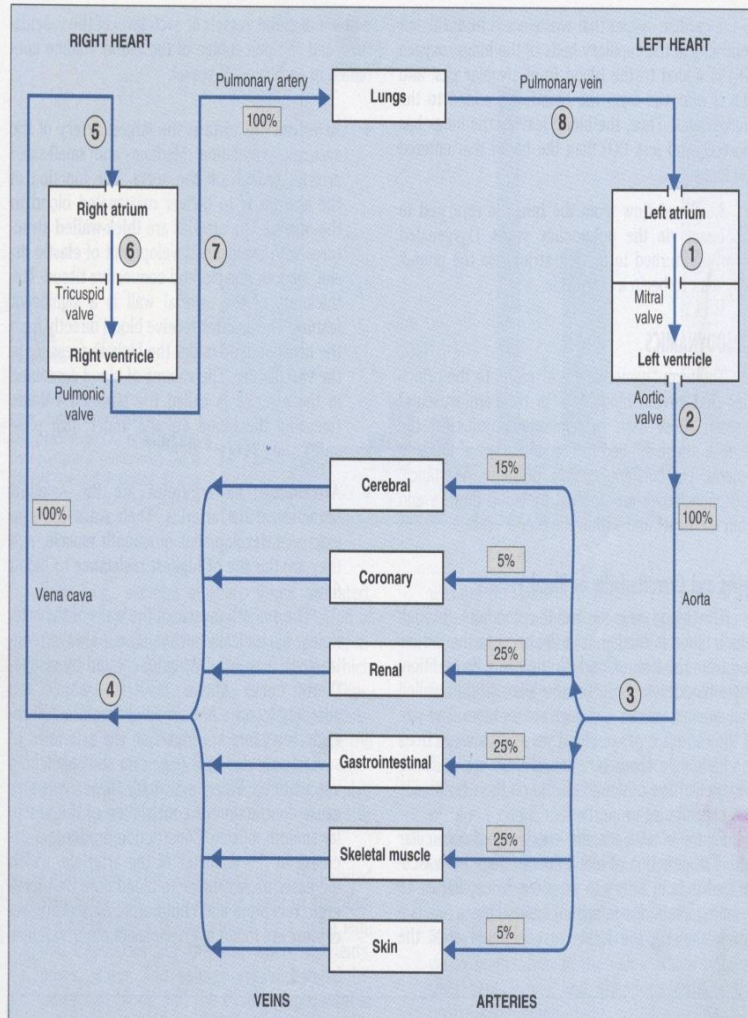


The LV pumps blood to the rest of the body
via the aorta (to systemic circulation)

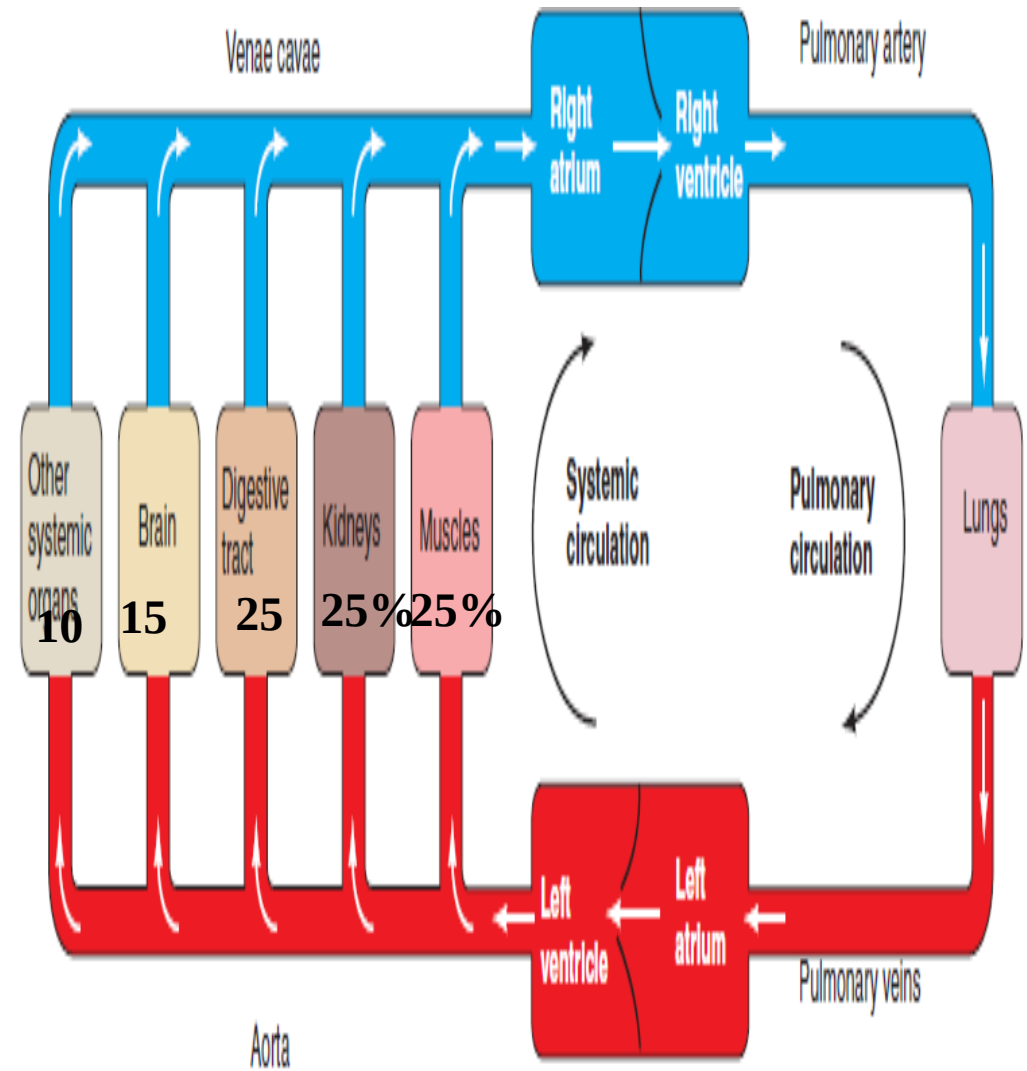
[Figure 1-6]



Pathway of Blood Through the Heart and Lungs

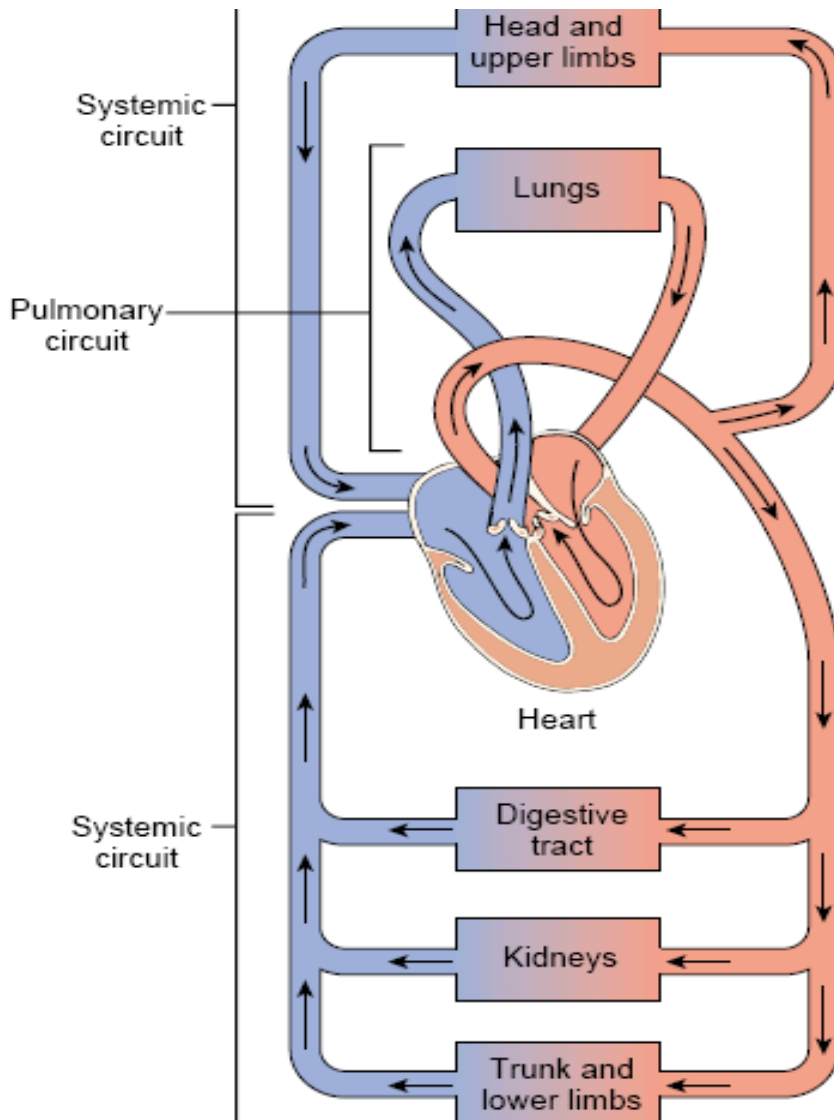


1

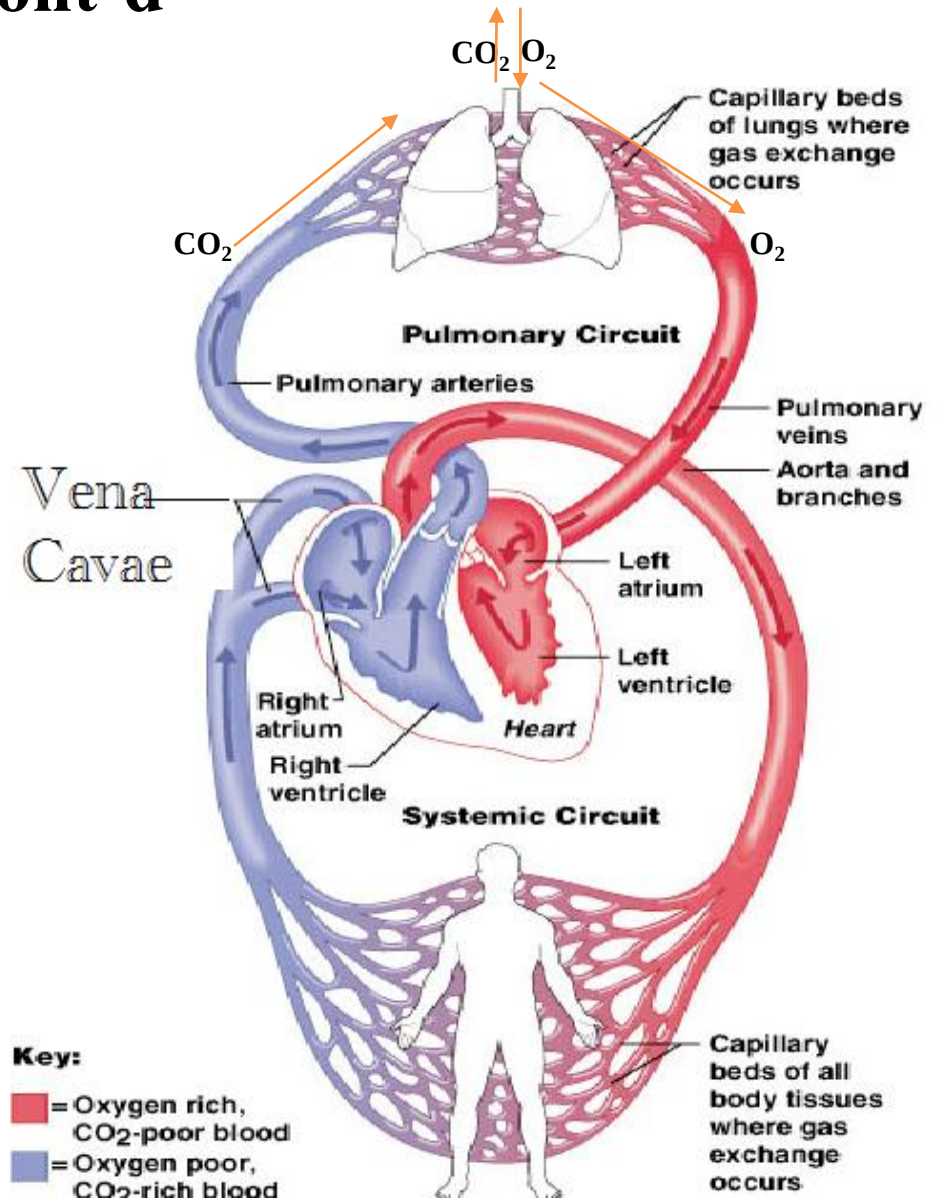


2

Cont'd

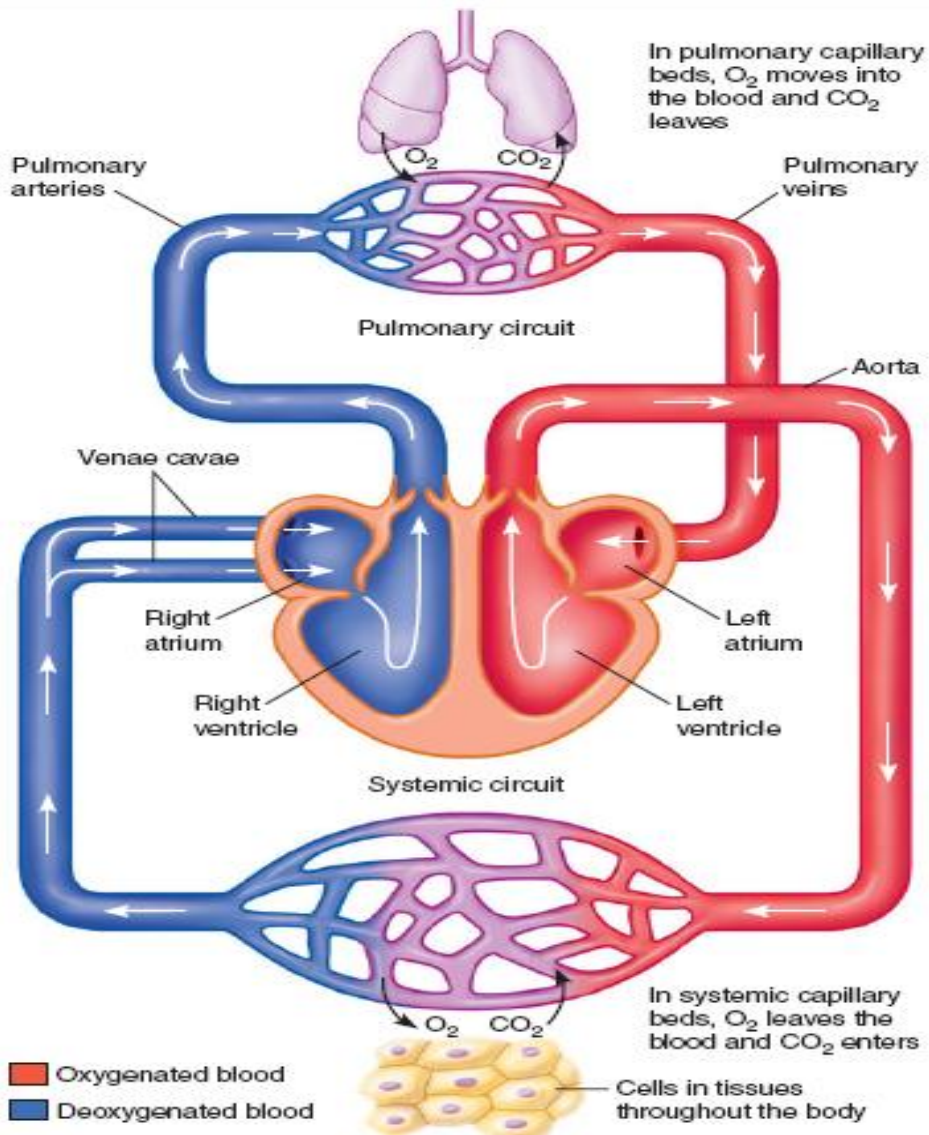


3

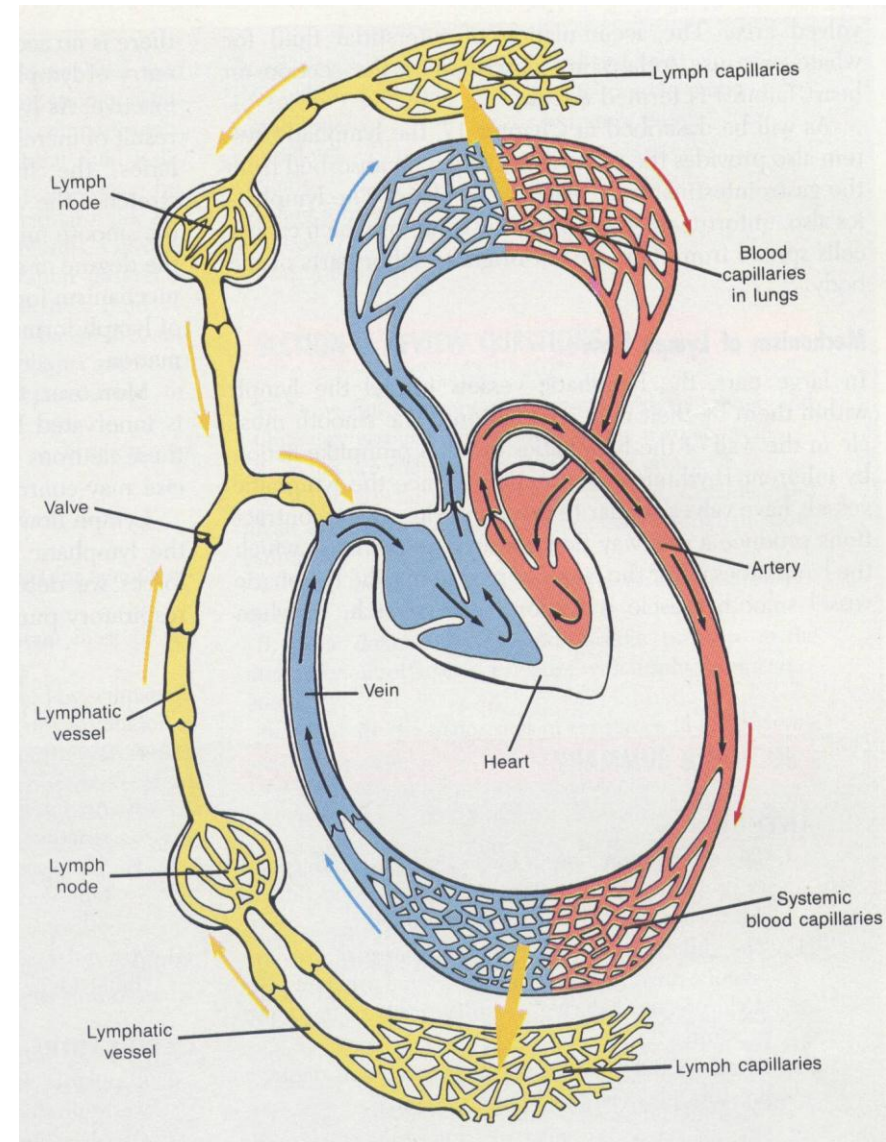


4

Cont'd

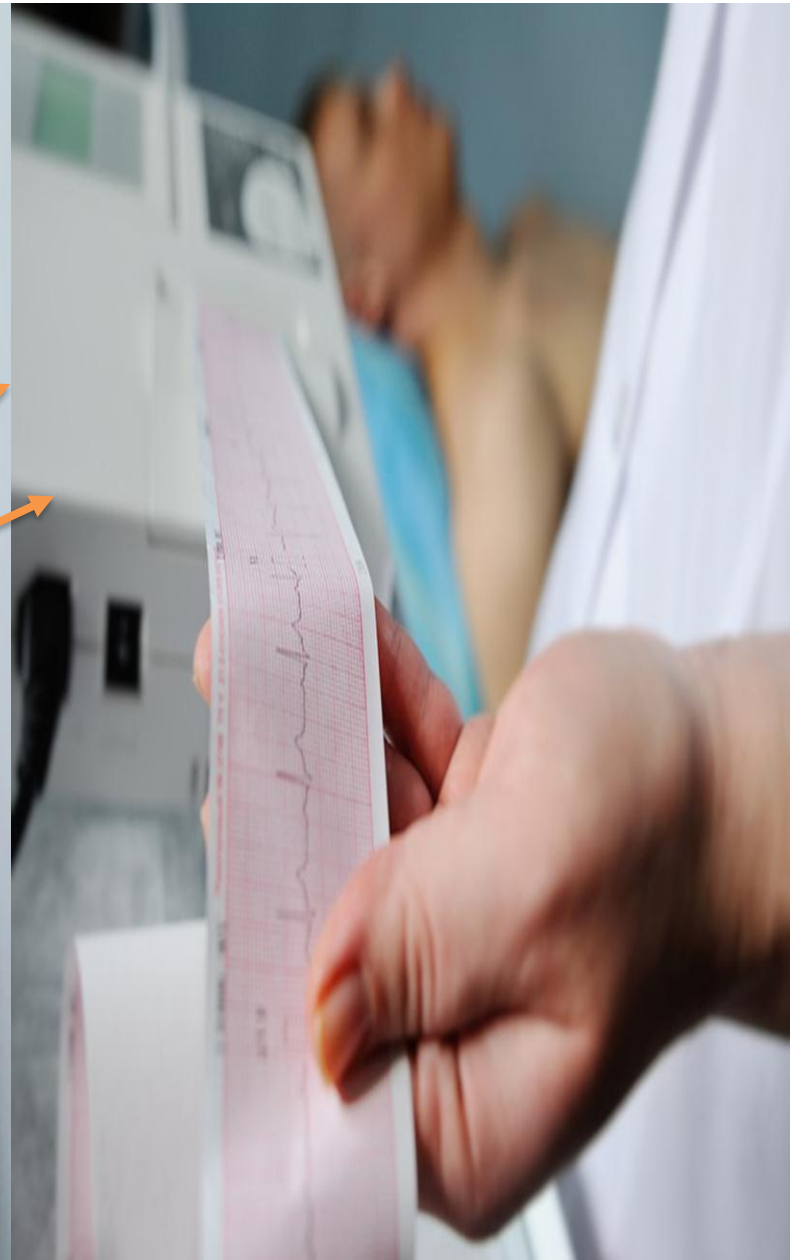
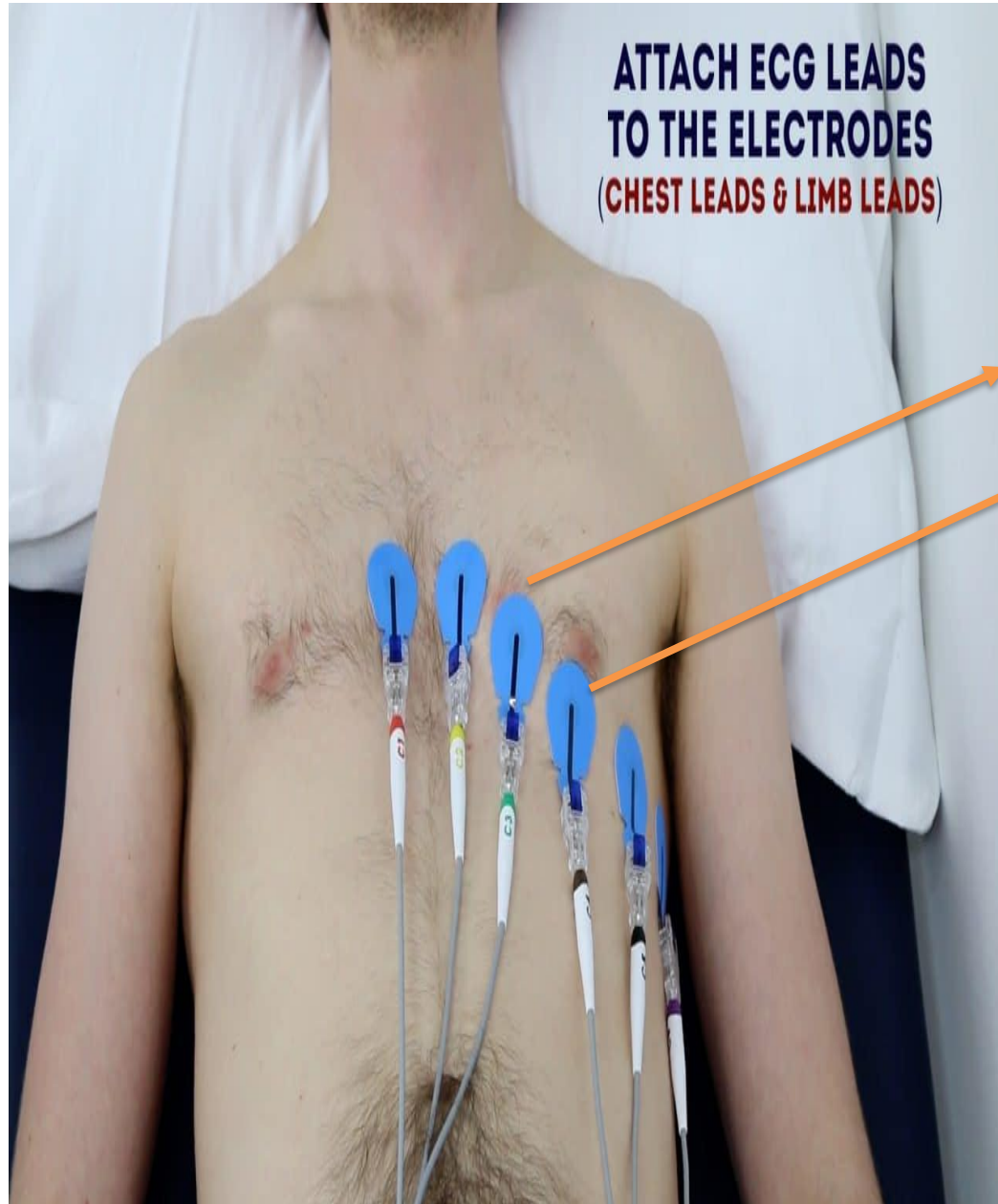


5



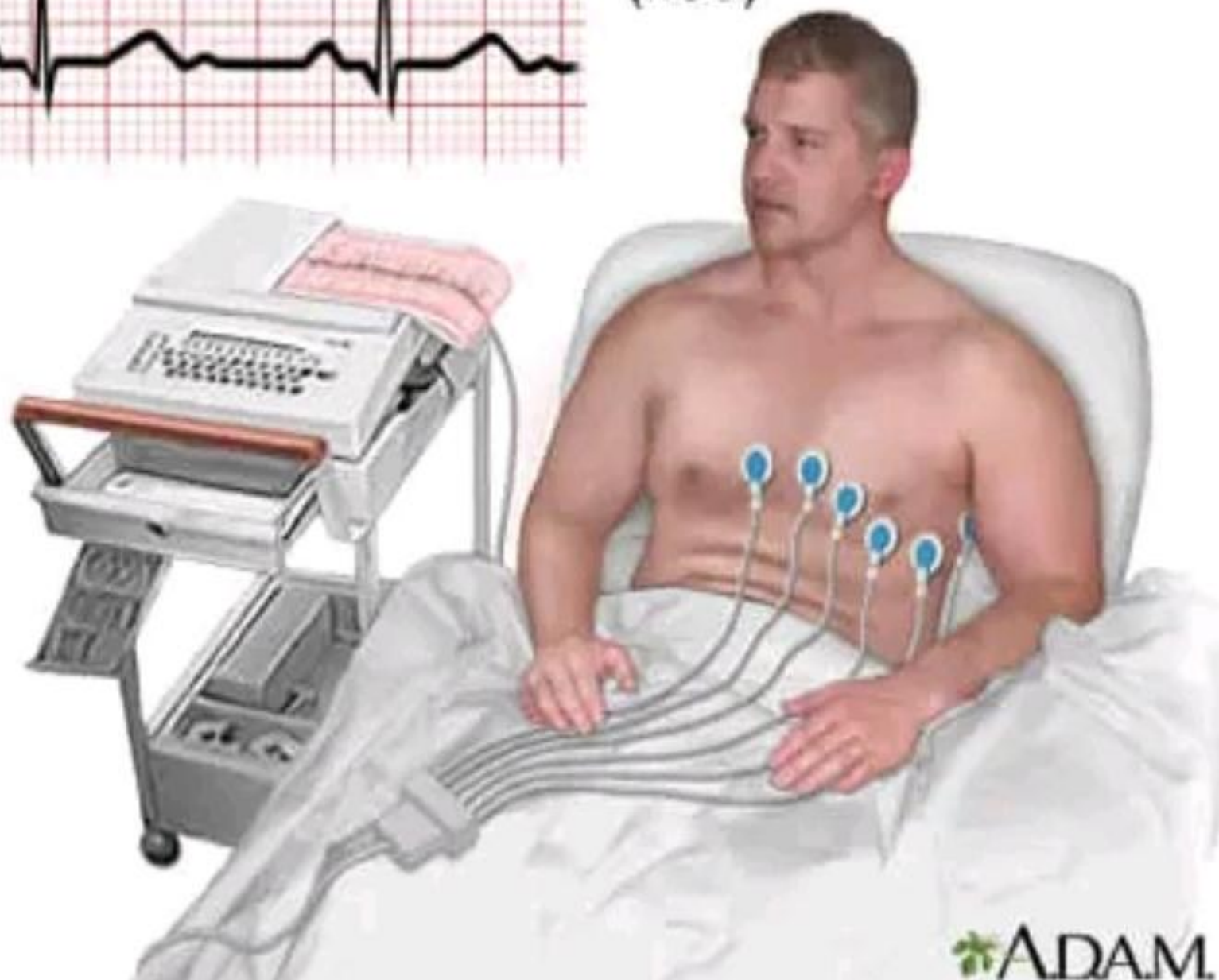
6

ECG/EKG





Electrocardiogram
(ECG)



Electrocardiography [ECG]

❑ Function of [Heart]

- ✓ Pumping blood for body's needs.

❑ Mechanical act of pumping blood is preceded by electrical stimulus/event.

- ✓ The different electrical event translate in to electrical cardiac vector then



- ✓ Translate into **ECG patterns** [W,S & I].

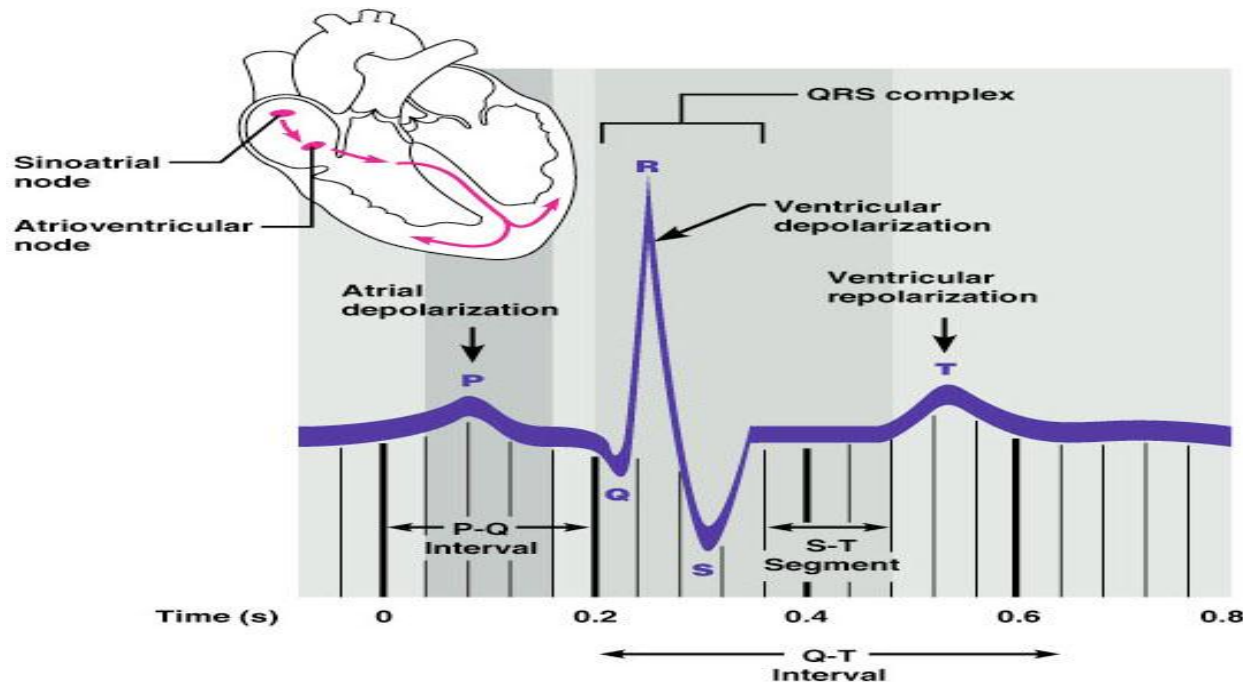
❑ **[ECG]** is a surface recording of these electrical activity of the heart from electrodes placed on the surface of the skin. because

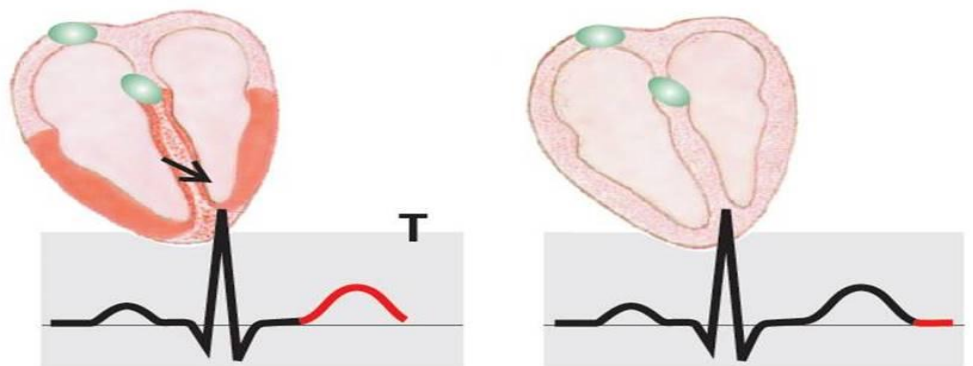
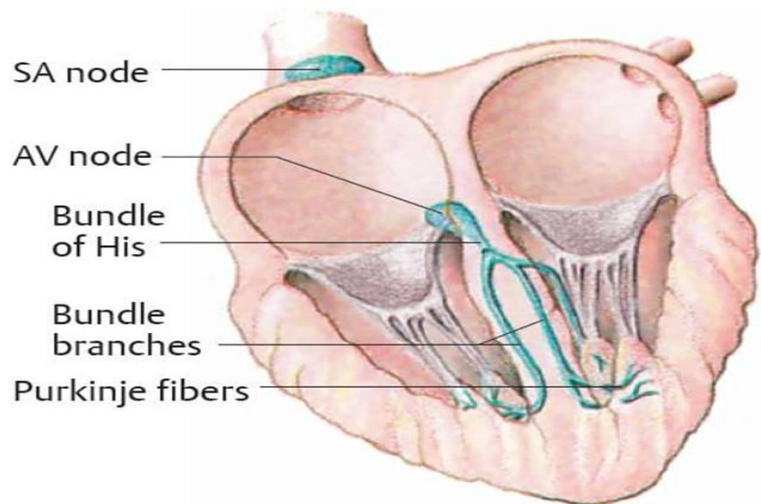
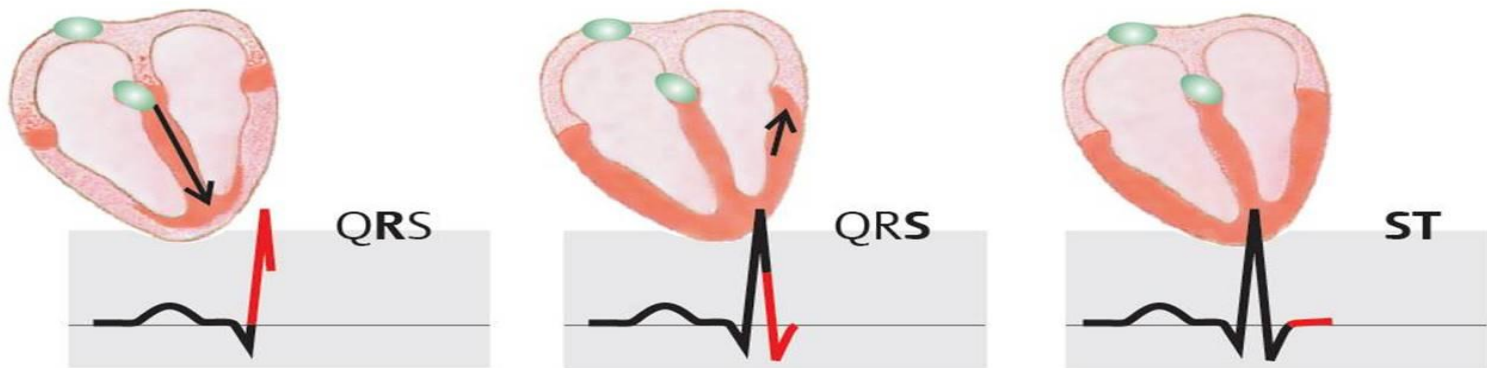
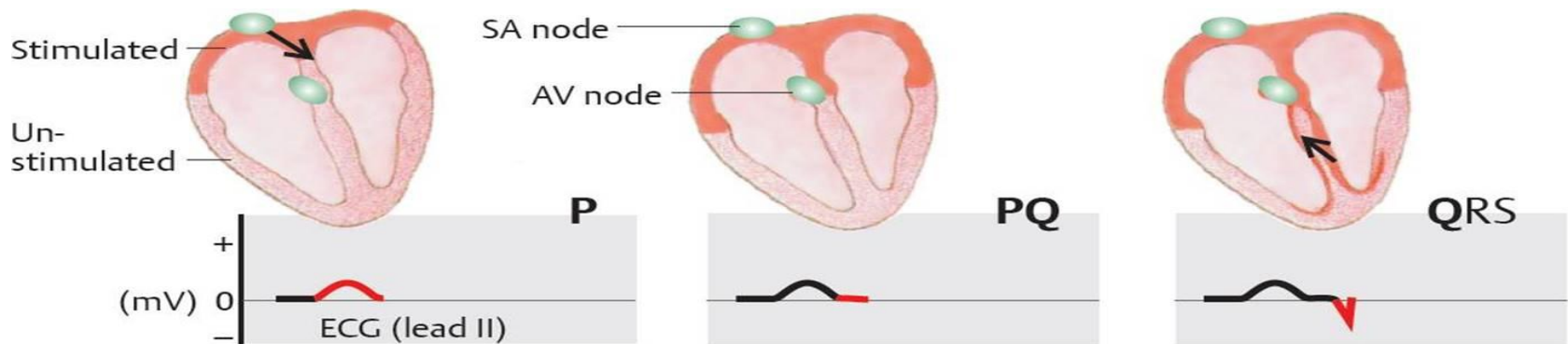
- ✓ The heart's electrical activity produces currents that radiate through the surrounding tissue & to the skin. i.e.
- ✓ When electrodes are attached to the skin, they sense these electrical currents then
- ✓ Transformed to an ECG monitor that represent the heart's depolarization-repolarization cycle.
- ✓ **ECG** is useful in diagnosing abnormal heart rates, arrhythmias, & damage of heart muscle.

Cont'd

❑ Electrical activity is recorded by Electrocardiograph

1. **P-wave** corresponds to depolarization of atria
2. **QRS complex** corresponds to depolarization of ventricle [vt]
 - ✓ **Q**=Septal part & **R**=Major part & **S**=basal part of Vt depolarization
3. **T wave** corresponds to ventricular repolarization
4. **Atrial repolarization** record is masked by the larger QRS complex





Characteristics of normal ECG

The ECG tracing consists of W,S,I

□ Waves [W]

- ✓ *P wave*
- ✓ *Q wave*
- ✓ *R wave*
- ✓ *S wave*
- ✓ *T wave*

□ Complex

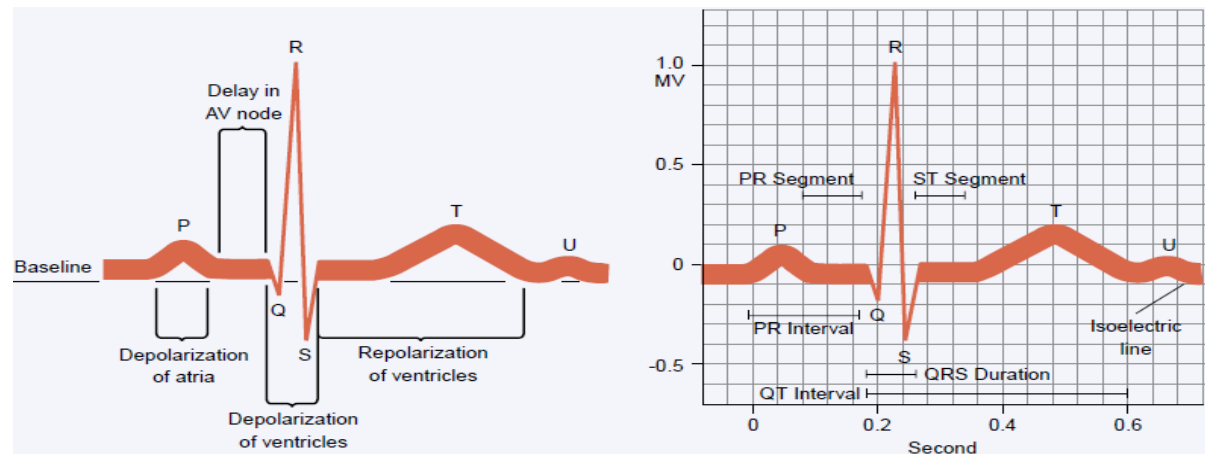
- ✓ *QRS*

□ Segments [S]

- ✓ *PR segment*
- ✓ *ST segment*

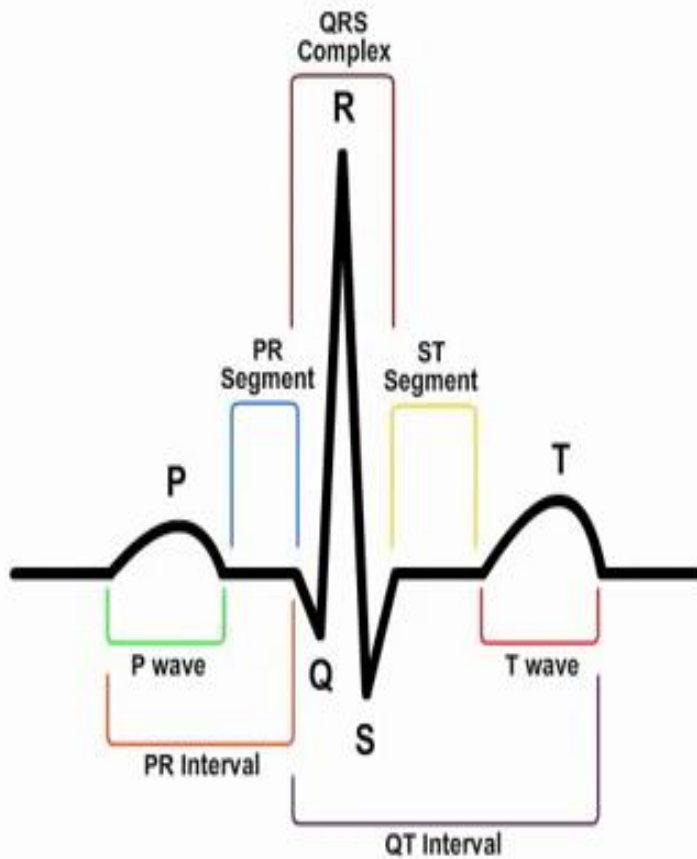
□ Intervals [I]

- ✓ *PR interval*
- ✓ *QT interval*

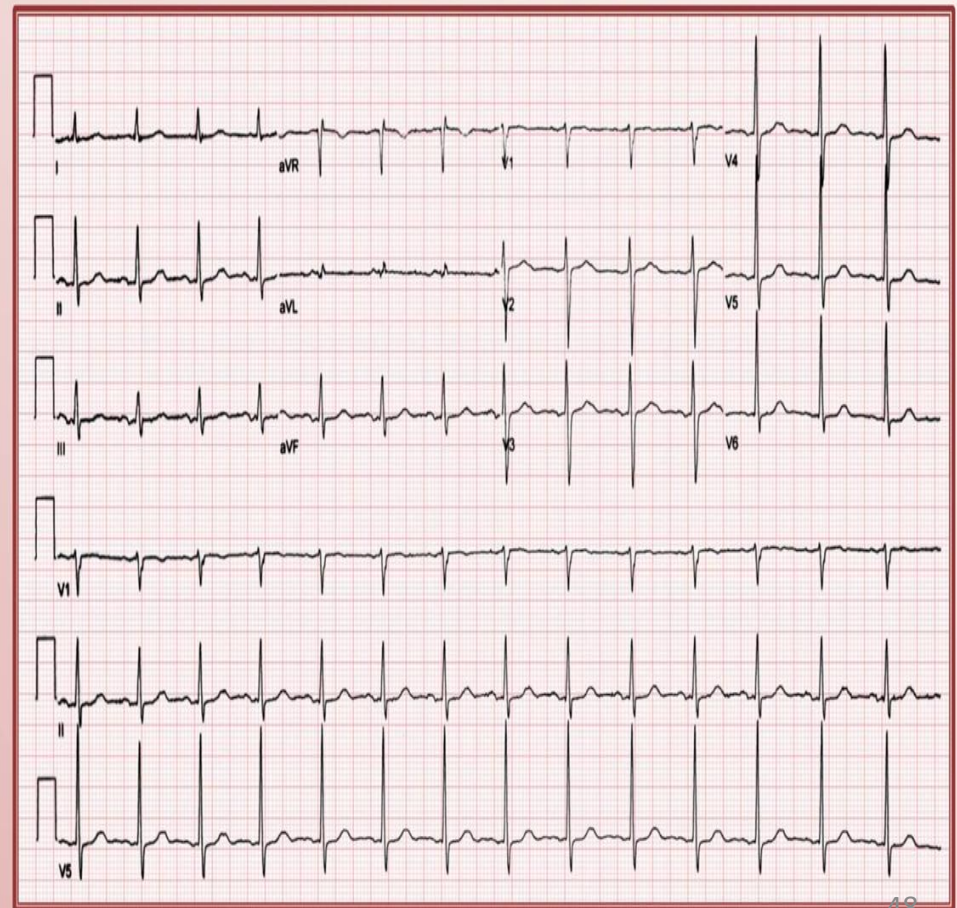


Cont'd

□ 1 cardiac cycle = 1 beat = 1 PQRST = systole + diastole



Normal EKG



ECG standard paper

❑ Consists of horizontal and vertical grid lines.

❖ The horizontal axis represents time.

✓ 300 big sq = 60 seconds

✓ 1big sq=5 small sqs=5mm

✓ The ECG =25mm/sec paper speed

✓ 1 big sq =0.2 second

✓ 5 small sq.=0.2 second

✓ 1 small sq.=0.04 second

❖ The vertical axis measures amplitude in 'mm' or in mV.

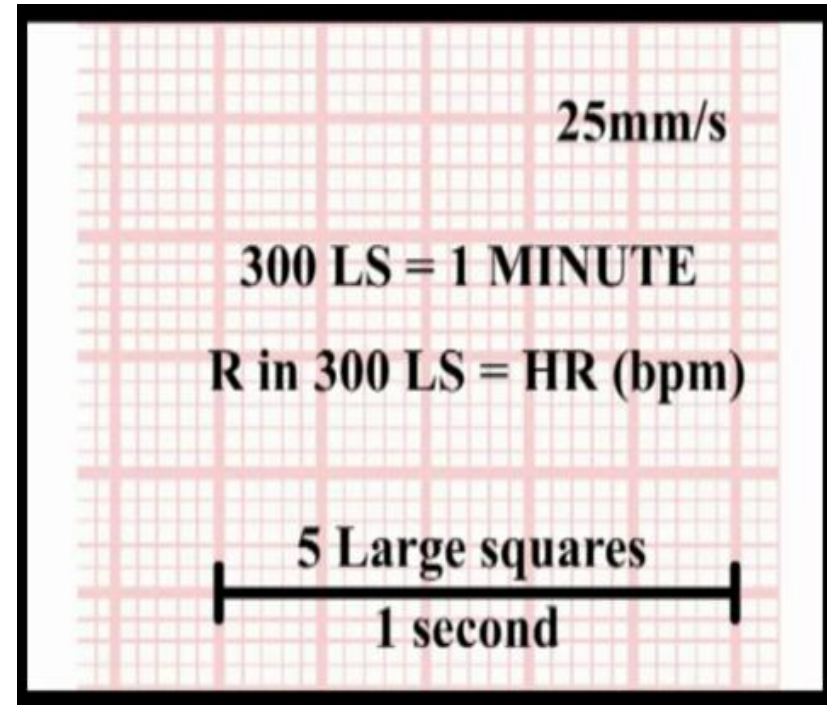
✓ Each small square represents 1mm = 0.1 mV.

✓ Each large square represents 5 mm = 0.5 mV.

❑ 1 big sq =5mm=0.2 sec

=5small sqs=0.04 sec

=1 small sq.=0.04 second



- Big square =Large square [LS]
- R=Q'R'S

Blood Vessels

❑ The network of tubular structure that carry (circulate) blood throughout the body.

1. **Arteries:** Carry O_2 rich blood away from the heart

✓ Deep seated

2. **Veins:** Carry CO_2 rich blood to the heart

✓ Superficial

3. **Capillaries:** Connect the arteries and veins

✓ Found in the terminal of the 2

❑ 62,000 miles (3 time the world) ?????

Blood Vessels: Classification

1. **Elastic vessels** : Aorta, big arteries
 - ✓ Pressure storing components
 - ✓ High ability of recoiling
2. **Resistance vessels** : Small arteries and arterioles
 - ✓ High muscular component
 - ✓ Develop high resistance
 - ✓ Regulate blood flow
3. **Exchange vessels:** Capillaries
 - ✓ Thin enough for exchange of subs
4. **Capacitance vessels:** Big to small veins
 - ✓ Very high capacity of distension
 - ✓ Can accommodate large volume of blood

Cont'd

1.Arteries:

- ❑ Transport (deliver) high-pressure oxygenated blood away from the heart to smaller arteries ➡ arterioles
 - ✓ **Thick**-walled, with extensive elastic tissue
 - ✓ Are under high pressure
 - ✓ Blood pressure averages 120 mm Hg/80 mm Hg
 - ✓ Aorta is the largest artery

2.Arterioles:

- ✓ The smallest branches of the 'arteries'
 - ✓ Connect arteries & capillaries
 - ✓ Site of the highest resistance in the CVS

Cont'd

3.Capillaries

- ✓ Smallest blood 'vessels'
- ✓ Are thin walled & consist of a single layer of endothelial cells
- ✓ Site of exchange of nutrients, water, & gases b/n **blood & cells or lungs.**
- ✓ Have the largest total cross-sectional or surface area.

☐ **Exchange occurs through**

- ✓ Mainly by simple diffusion

Cont'd

❑ Diffusion via capillaries enhanced by:

- ✓ Thin capillary wall (just one cell thick)
- ✓ Large number
 - ✓ High surface area through which exchange can occur
- ✓ Relatively slow flow of blood (providing more time for exchange to occur)

4. Venules: A very small vein

- ✓ Connect capillaries & veins & collect blood from capillaries

5. Veins : Are **thin** tubes , return blood to heart

- ✓ Act as reservoir of blood & transport “low pressure” blood
 - ✓ Serve as low-pressure passageway
- ✓ Vena cava is the largest vein

1

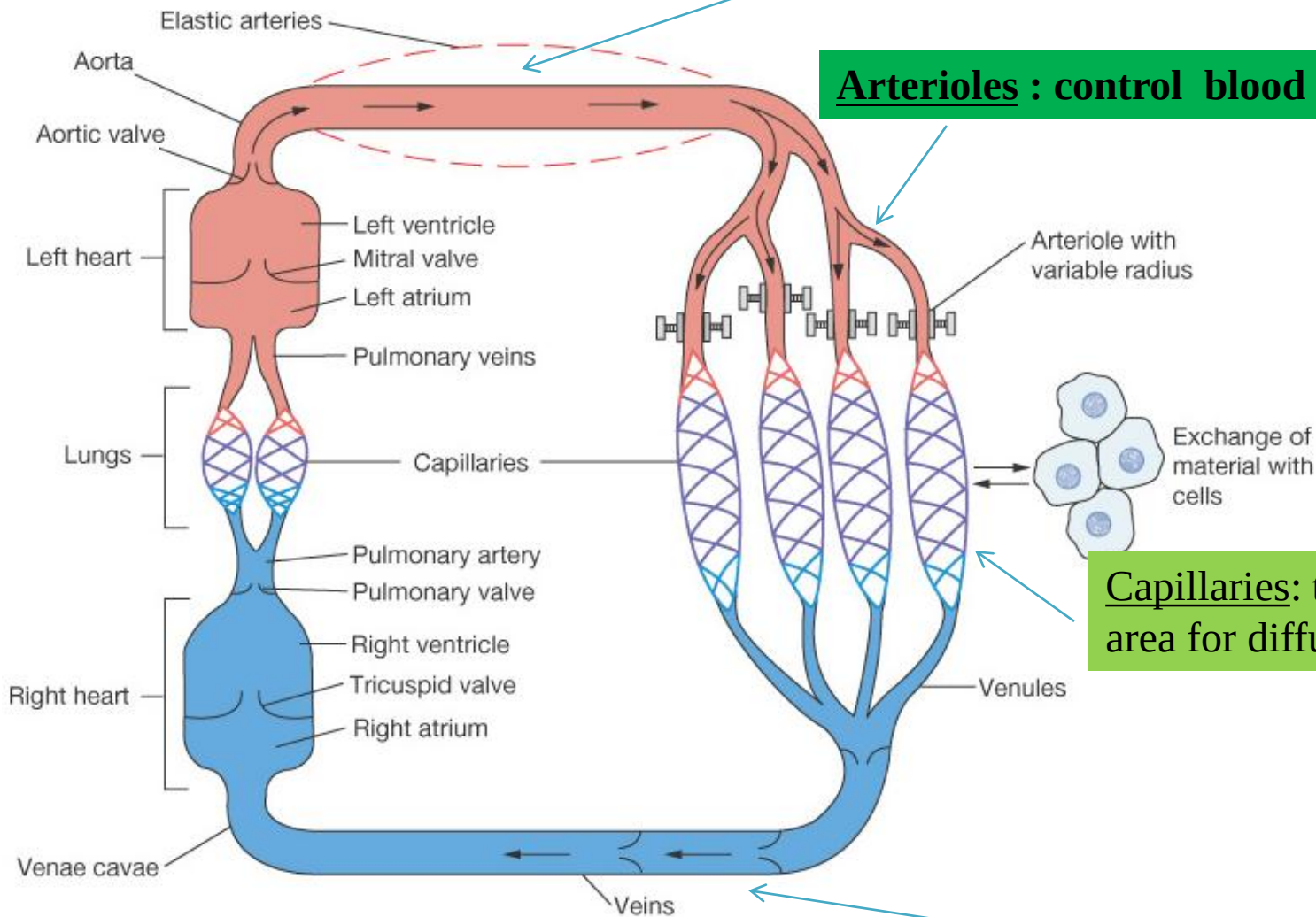
Cont'd

Strong and elastic arteries

Arterioles : control blood flow and pressure

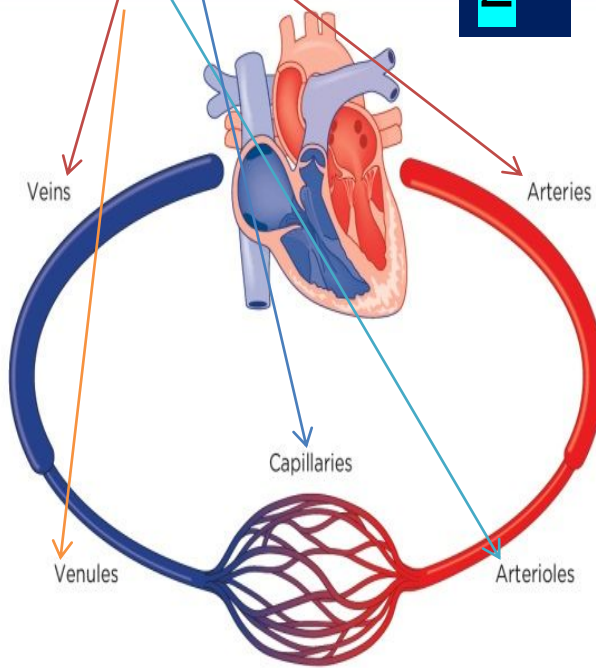
Capillaries: thin and with large area for diffusional exchange

Veins: Compliant [not resistant], large, low resistance
-Have valves & assure blood return to the heart



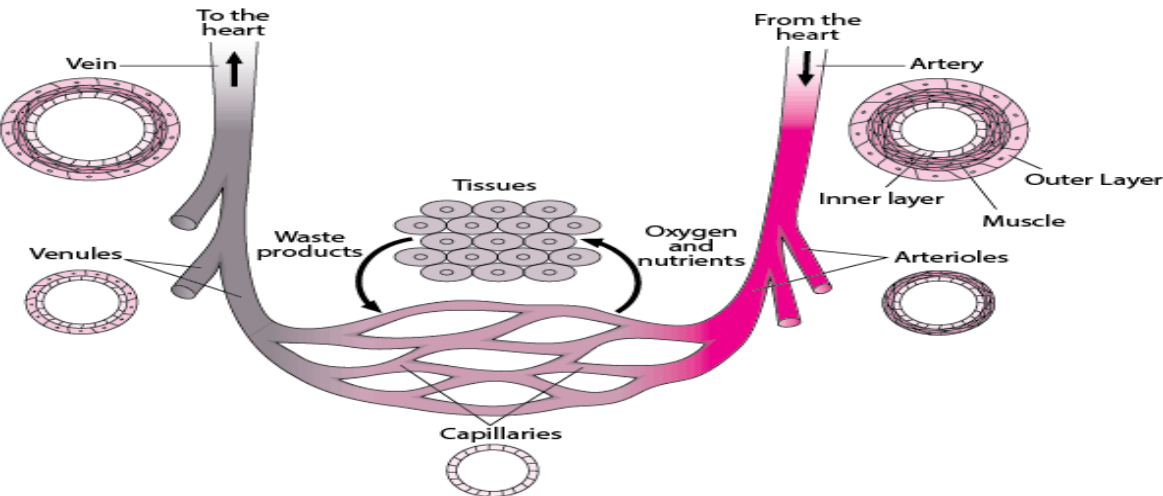
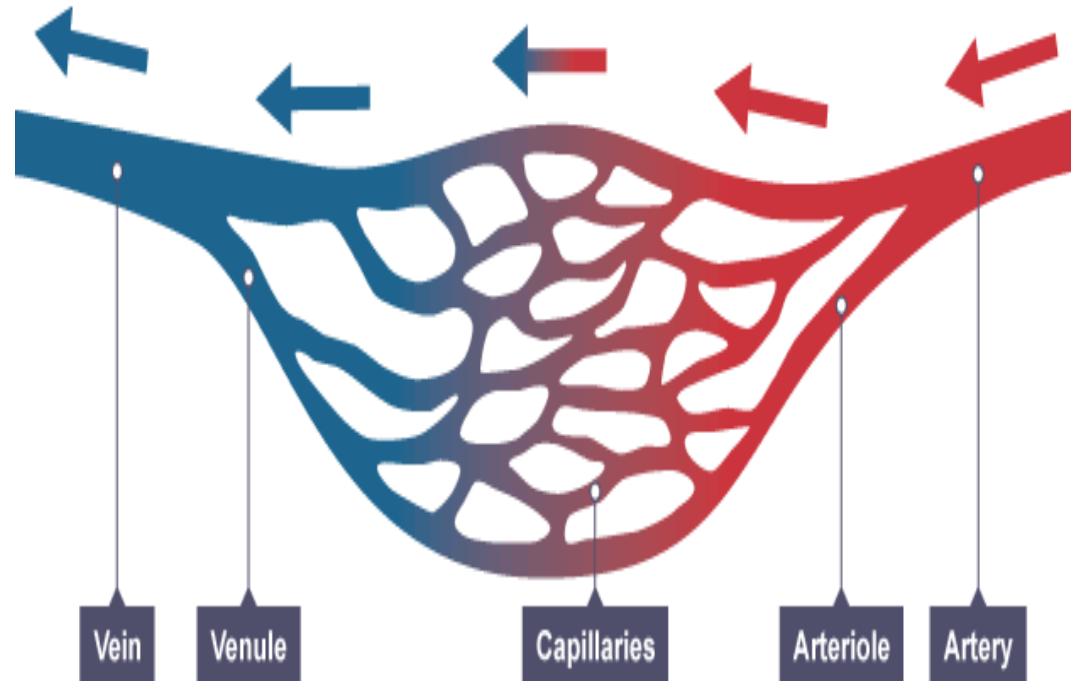
types of blood vessels

2



Cont'd

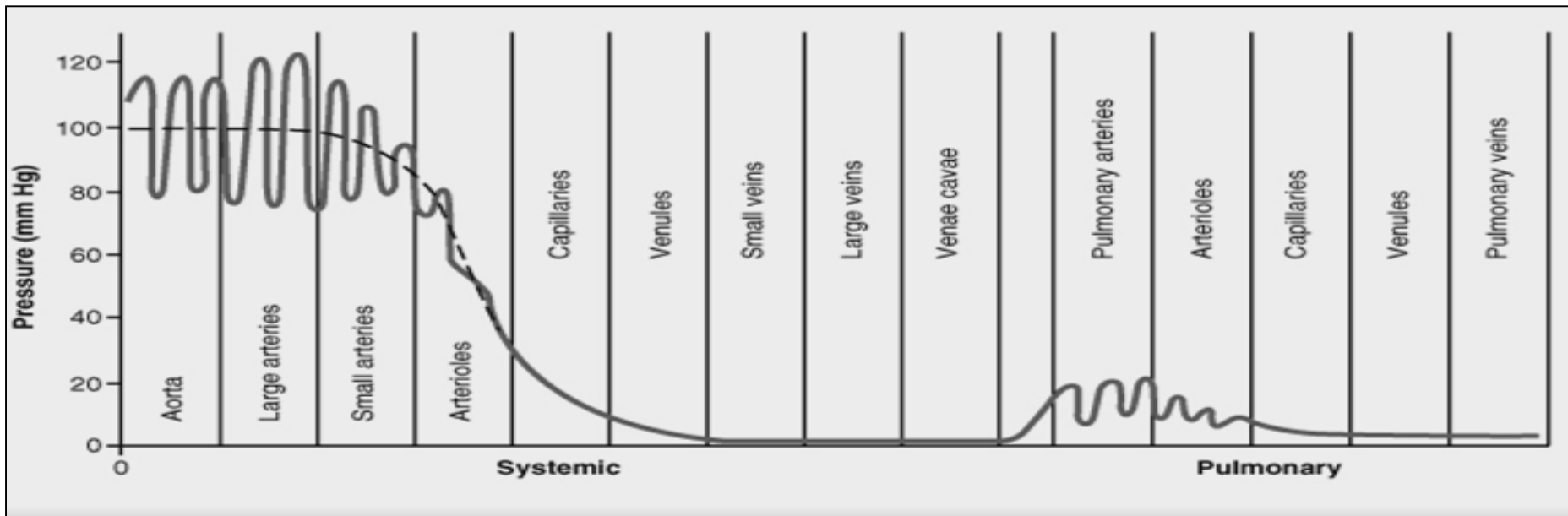
3



4

Cont'd

	Arteries	Veins	Capillaries
1.Functions	Send blood from heart=Distributer	Send blood to heart =Returner	Allow for material exchange with tissues [cells]=Supplier
2.Pressure	High	Low [less resistant to BF]	Low
3.Lumen diameter [size]	Narrow	Wide	Extremely narrow
4.Structure of the Wall	Thick	Thin	Extremely thin [one cell thick]
5.Muscle & elastic fibers	Large amounts [muscular]	Small amounts	None
6.Valves	No	Yes [prevent back flow]	No
7.Blood flow [BF]	V . Fast	Slow	V. Slow



Blood Supply to the Heart

❑ The coronary arteries (CA)

❑ Either of 2 arteries that originate in the aorta and supply the heart muscle with blood.

✓ CA are branches of ascending aorta

❑ Coronary veins that receives CA blood and empties into

✓ Coronary sinus → the right atrium.

❑ **Resting coronary blood flow = 250 ml/min = 5% of CO**

❑ Coronary arterial **diseases** (Angina pectoris)

✓ Ischemia = [low blood supply to the heart muscle cells] → Myocardial Infarction [heart cell death → destruction of heart tissue]

Arterial Blood Pressure [ABP]

- ❑ Is force or pressure pushing against the wall of the arteries. OR
 - ✓ Is the lateral pressure exerted by the flowing blood on the wall of the arteries.
 - ✓ It has 2 components: **systolic** and **diastolic** pressure
- ❑ **ABP** is measured in millimeters of mercury [**mmHg**].
- ❑ Always recorded as 2 numbers, such as **120/80**
 - ✓ Where the top number is the **systolic pressure** and
 - ✓ The bottom number is the **diastolic BP**.

Cont'd

Systolic Blood Pressure [S]



- ❑ Is the Max 'P' exerted by the blood against the artery's wall.
 - ✓ Is peak pressure [P].
 - ✓ It results when the **ventricles contract** and pump blood
 - ✓ Normal= ~ 120 mmHg [S]

Diastolic Blood Pressure [D]



- ❑ Is the Min 'P' on the wall of arteries when the heart is at rest [relaxed]
- ❑ Is the **lowest pressure** in the artery.
 - ✓ It results when the **ventricles relaxed** and
 - ✓ Normal= ~80 mmHg [D]

Cont'd

- The **pulse pressure** is the difference b/n the measured S & D 'P'.


$$\checkmark P_{\text{pulse}} = P_{\text{systole}} - P_{\text{diastole}}$$

- **MAP** → it is the average of S & D pressures through the cardiac cycle with respect to time.


$$\square \mathbf{MAP} = D + (S-D)/3$$

- If S=120 & D=80, **Find MAP?**

$$\checkmark \mathbf{MAP} = D + (S-D)/3 = (2D+S)/3 = 80 + (120-80)/3 = \mathbf{93mmHg}$$

$$\checkmark P_{\text{pulse}} = 120-80 = \mathbf{40mmHg}$$

Methods of Measurement

Clinically BP can be measured by:

A. Direct method (**invasive**)

✓ Intra-arterial

B. Indirect method (**non-invasive**)

1. Oscillometric method (Automated)

2. Auscultatory method (manual)

Cont'd

A. Direct (invasive) method

❑ It measures ABP internally .

✓ By using a sensitive IV catheter inserted into the artery. But

❑ Due to its invasiveness and its inability to be applied to large number of patients

✓ This method is not routinely practical.

B. Indirect (non-invasive) method

1. Oscillometric method (Automated)

✓ The cuff fitted on the upper arm .And

✓ Inflated and released by an electrically operated pump and valve.

✓ Uses an electronic pressure sensor with

✓ A numerical readout of BP.



Cont'd

2. Auscultation (manual) →→ Hearing Korotkof sound



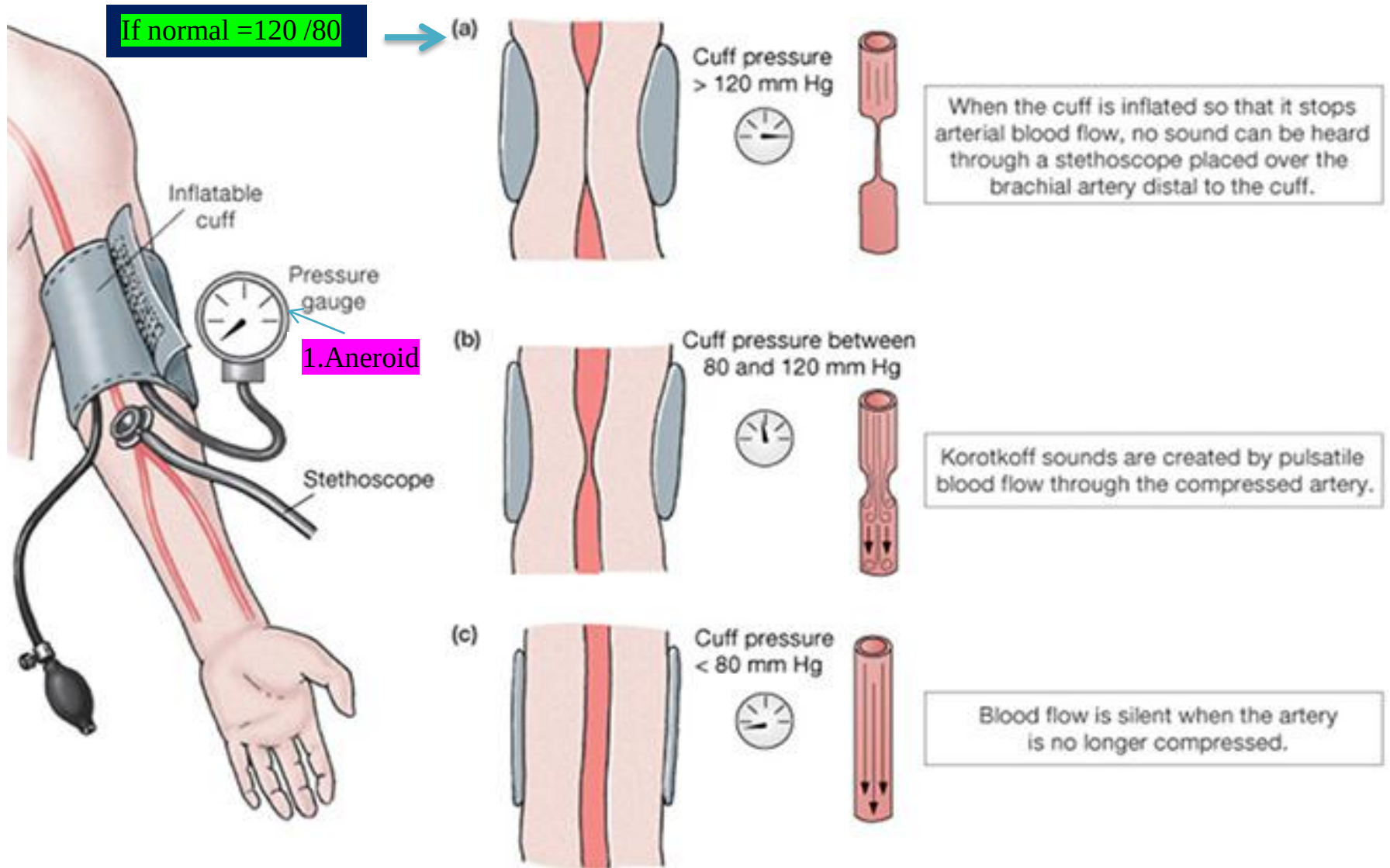
- ❑ Is Auscultatoric measurement
- ❑ This method determine BP by using the sphygmomanometer and stethoscope .

1.Sphygmomanometer :is an instrument that measures BP which consists of.

- ✓ A cuff
- ✓ Inflating bulb & valve
- ✓ Pressure gauge

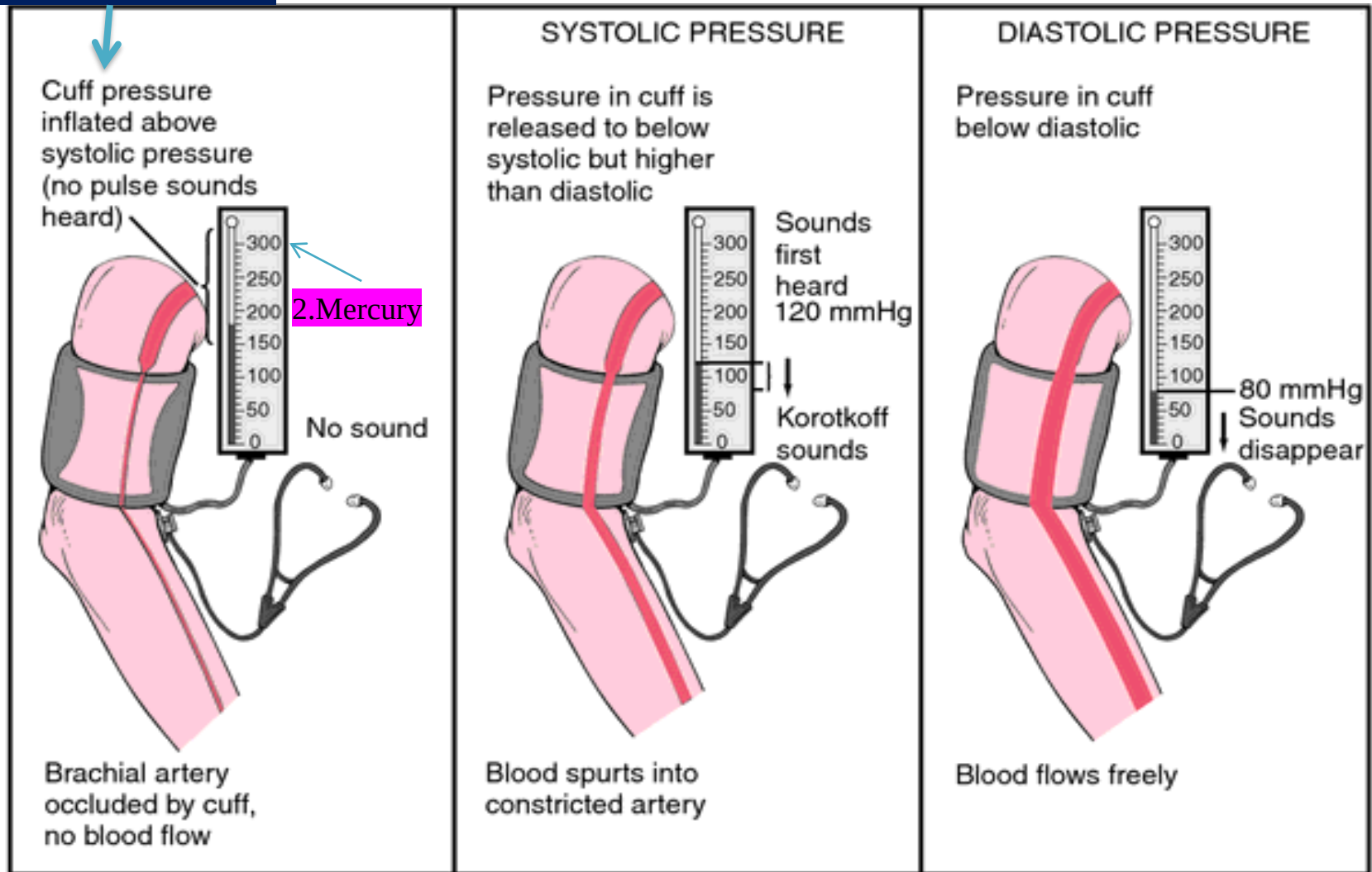
2.Stethoscope : is an instrument used to hear sound

Measurement of “ABP” by Auscultatory method

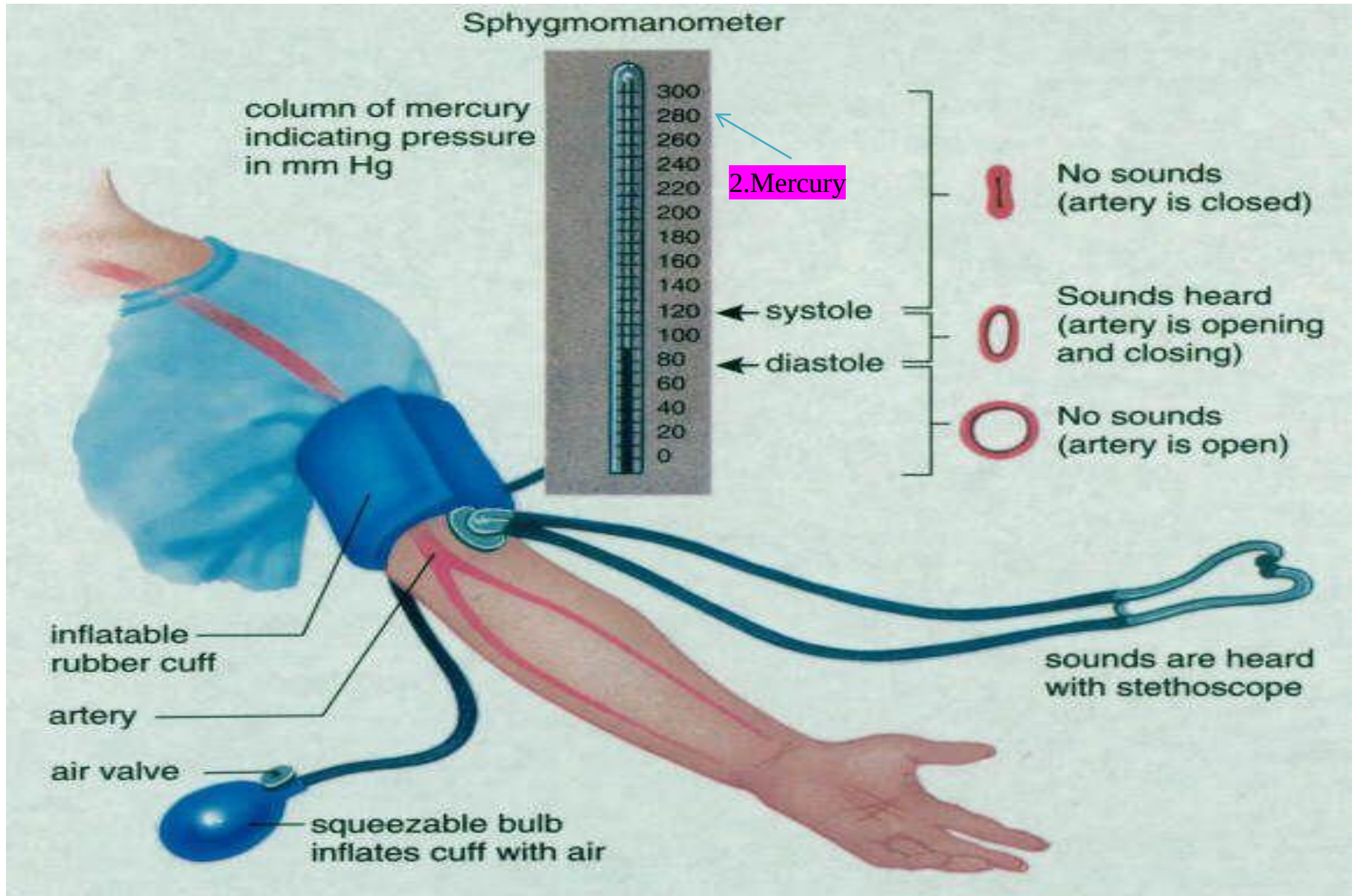


Cont'd

If normal = 120 / 80



Cont'd



Summary [ABP]

□ **ABP** is a force exerted by the blood on the wall of the blood vessels (arteries)

✓ Has 2 components: **systolic** and **diastolic** pressure

✓ Expressed as : BP=systolic/diastolic

✓ Normal : SBP: 100-130mmHg →→120 mm Hg

:DBP: 70-90mmHg →→ 80 mm Hg

✓ $P_P = S_{BP} - D_{BP}$

✓ $MAP = P_D + 1/3(P_S - P_D)$

✓ **$ABP = CO \times TPR$**

✓ **$CO = HR \times SV$**

□ **TPR**=Total peripheral resistance

□ The average BP for normal person is 120/80 mmHg

What Factors affecting Or Determine BP ?

1. Cardiac output [CO]
2. Peripheral vascular resistance (PVR)
3. Blood volume [BV]
4. Age: ABP rises as age increases
5. Time of the day: ↑ morning time [E & NE]
6. Body wt: ABP higher by 10-15 mmHg in obese
7. Gravity: every 1 cm distance below the level of the heart ↑ pressure by 0.77 mm Hg
8. Exercise: increases ABP by 40 – 50 mm Hg
9. Emotion: increases ABP by 10 – 30 mm Hg
10. Deep sleep: decreases ABP by 20 mm Hg

Hypertension (HT)

- ❑ **Hypertension**: A sustained increase in BP ($\geq 140/90$ mm Hg) [on repeated BP measurement].
- ❑ **Criteria for HTN in Adults**

Classification (stages) of HT for adults

Category	Systolic (mmHg)	Diastolic (mmHg)
Hypotension		
Normal	120	80
Pre-HT	120–139	80–89
HT :stage 1	140–159	90-99
HT: stage 2	160–179	100–109
HT: stage 3 (Hypertensive emergency, crisis)	≥ 180	≥ 110

Etiology [Types] of HT

- ❑ **Silent killer** = no symptoms till vital organs are damaged [most people unaware of the problem bcz it may have no warning sign & symptoms].

Two types of HT based on etiology

1. Primary (essential) HT [90%] :

- ✓ Unknown etiology OR no identifiable cause
- ✓ No established causes known = idiopathic HT

2. Secondary HT [10%]:

- ✓ Known cause OR [occurs due to some underlying disease] such as ;
 - ✓ Kidney diseases [Renal artery stenosis & excessive release of rennin]
 - ✓ **Phaeochromocytoma** (Tumor of the adrenal medulla)
 - ✓ Excess glucocorticoids: water reabsorption ,pregnancy etc

Major Risk factors

- ✓ Smoking, diabetes mellitus, age > 60yrs,
- ✓ Sex (men and postmenopausal women)
- ✓ Dietary factors [high salt or fat intake], dyslipidemia, stress , obesity (overweight)
- ✓ Having a sedentary lifestyle (lack of physical activity), family history of cardiovascular disease [genetic].

Regulation of BP [how body regulate]

Two basic mechanisms for regulating BP

1.A short-term: nervous or neural mechanism works with in few seconds to-minutes)

- ✓ Is controlled by the autonomic nervous system (ANS).
- ✓ Changes in **BP** are detected by baroreceptors OR chemoreceptors
- ✓ Baroreceptors are sensitive to change in pressure
 - ✓ Eventually, change or regulate blood vessel diameter ,HR (contractility) ,peripheral resistance, stroke volume & change BP

Baroreceptors [pressure sensor (e.g. BP)] OR chemoreceptor [chemical sensor (e.g.O₂ & CO₂)]

Cont'd

❑ Short-term changes in BP are mediated by the autonomic nervous system.

- ✓ This change is **sensed by** baroreceptors (located in the aortic arch and carotid artery) then
- ✓ **Processed by** the **Vasomotor Centers in the Medulla [VCM]**
- ✓ **VCM** regulates sympathetic and parasympathetic nervous system output.

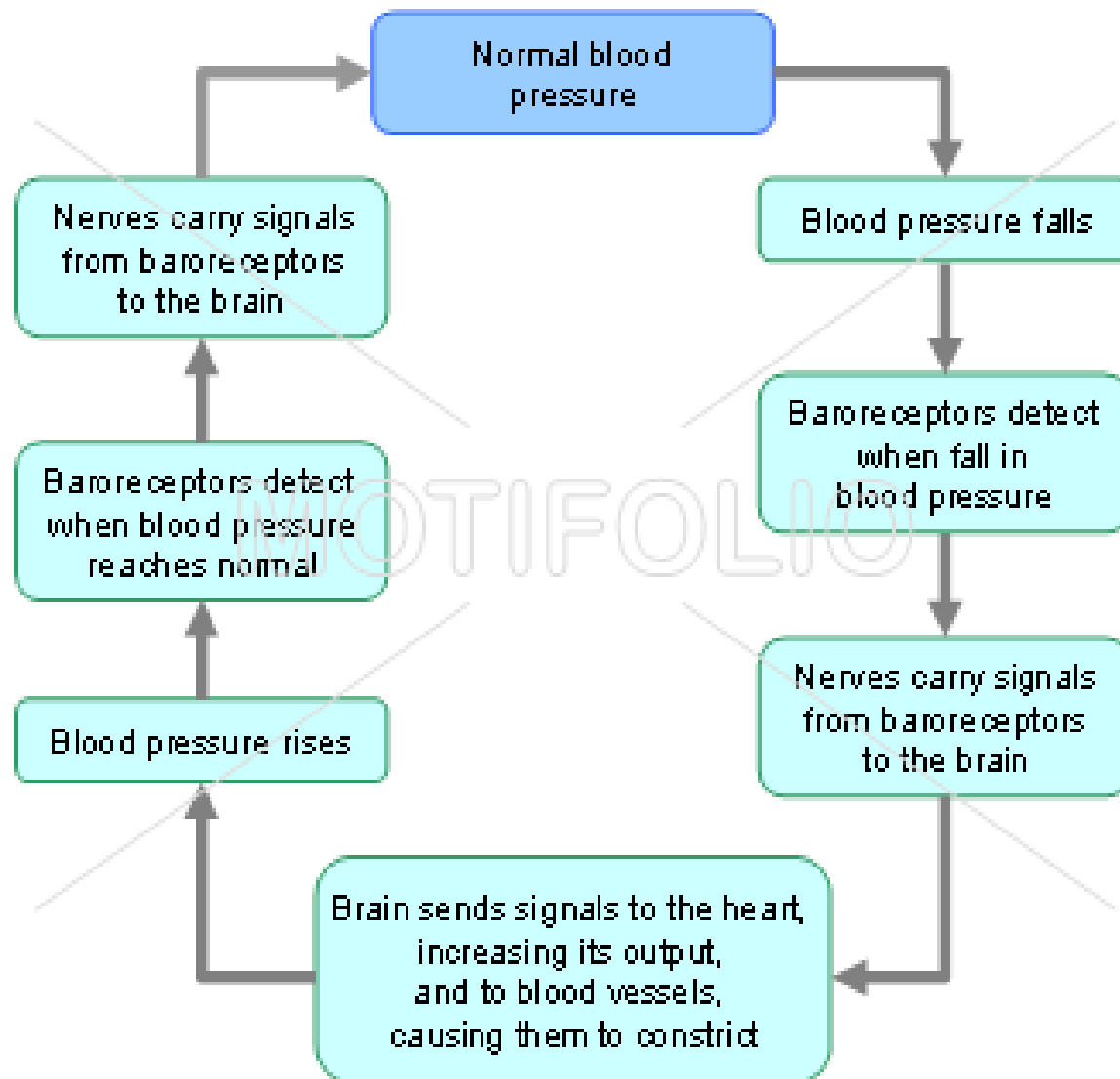
For example

❑ If a drop [↓↓] in BP is **detected** by the baroreceptor

- ✓ The change **send** to the brain [**VCM**]
- ✓ Resulting in an **increase in sympathetic** tone and a decrease in parasympathetic.
- ✓ **The final out put gives rise** to an **increased** in vascular resistance, venous tone, HR, CO, contractility of the myocardium.

Cont'd

Short-term control of blood pressure



To sum up short-term

Cont'd

❑ Falling BP



❑ Baroreceptors detect & send to the brain



❑ Brain ↓PS activity & ↑Sympathetic activity



❑ Effects

1. Heart

-↑ HR & contractility of the myocardium.

2. Vessels

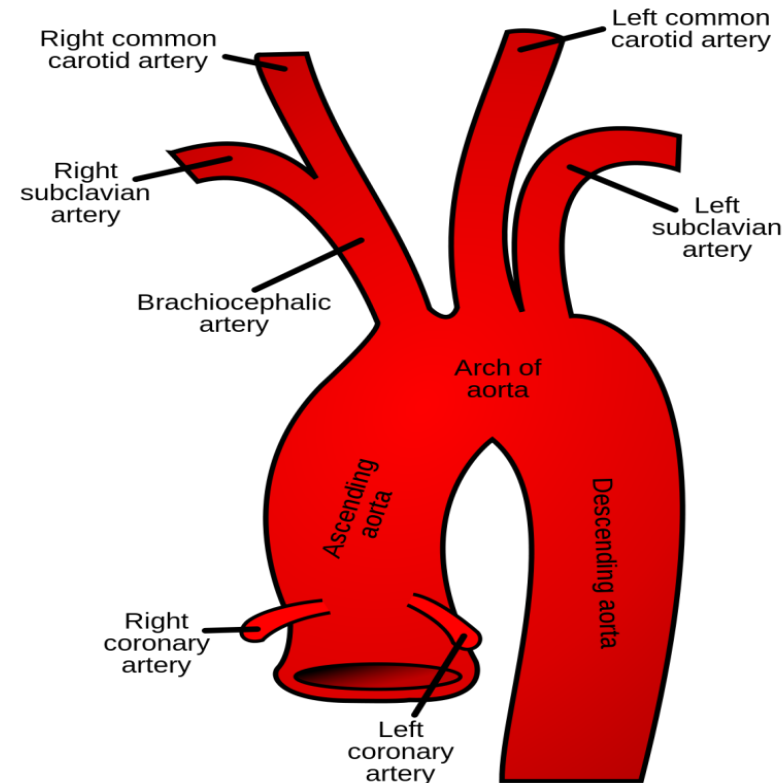
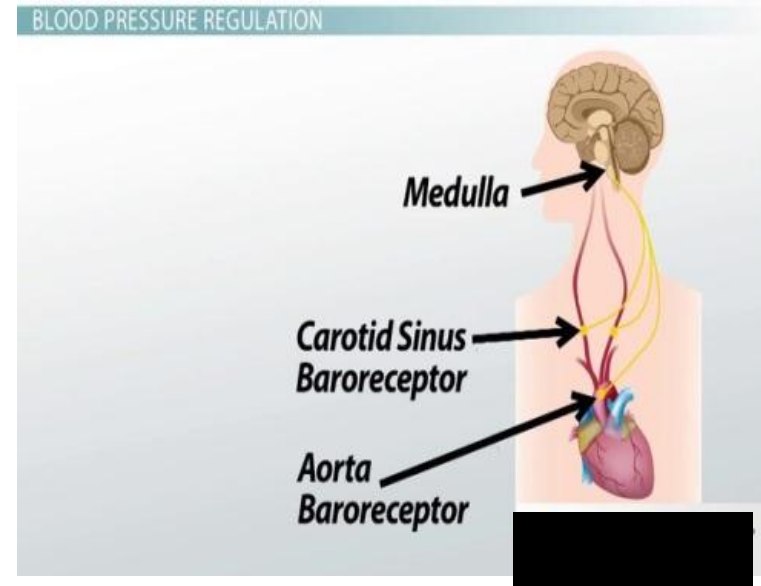
-↑ vasoconstriction

3. Adrenal gland

-Release NE → which ↑ HR



↑ ABP



Cont'd

2.Long-term :Renal or hormonal control works within few days to-wks basis

- ☐ Is primarily accomplished by altering or regulating blood volume
 - ✓ The loss of blood by hemorrhage or accident will lower BP & trigger a process to restore blood volume →→BP will back to normal
- ☐ Is Renal body fluid control system
 - ✓ By kidney via **RAAS** system
 - ✓ Renin
 - ✓ AgII
 - ✓ Aldosterone
 - ✓ ADH (vasopressin)

Cont'd

❑ A drop in renal blood flow or BP results in an increase in **renin** release from the kidney.



✓ An increase in **renin** release leads to



✓ An increase in circulating **angiotensin II**.



❑ The primary actions of **angiotensin II** are:

1. To stimulate the synthesis and secretion of aldosterone.

✓ Aldosterone acts on the kidney to retain Na^+ (and therefore water), leading to an increase in blood volume.

2. To raise blood pressure by direct vasoconstrictor effects (angiotensin II is one of the most potent vasoconstrictor known).

3. Stimulates the release of vasopressin (antidiuretic hormone, ADH) -Which acts upon the kidneys to increase fluid retention

4. Also stimulates thirst centers within the brain leading to high fluid intake.



Cont'd

1

Hypotension
Decreased Sodium Delivery ↓[BP]

Kidney

Renin

Angiotensinogen

AI

ACE

Angiotensin II (AII)

Adrenal Cortex

Cardiac & Vascular Hypertrophy

Pituitary

Aldosterone

Systemic Vasoconstriction

Thirst

ADH

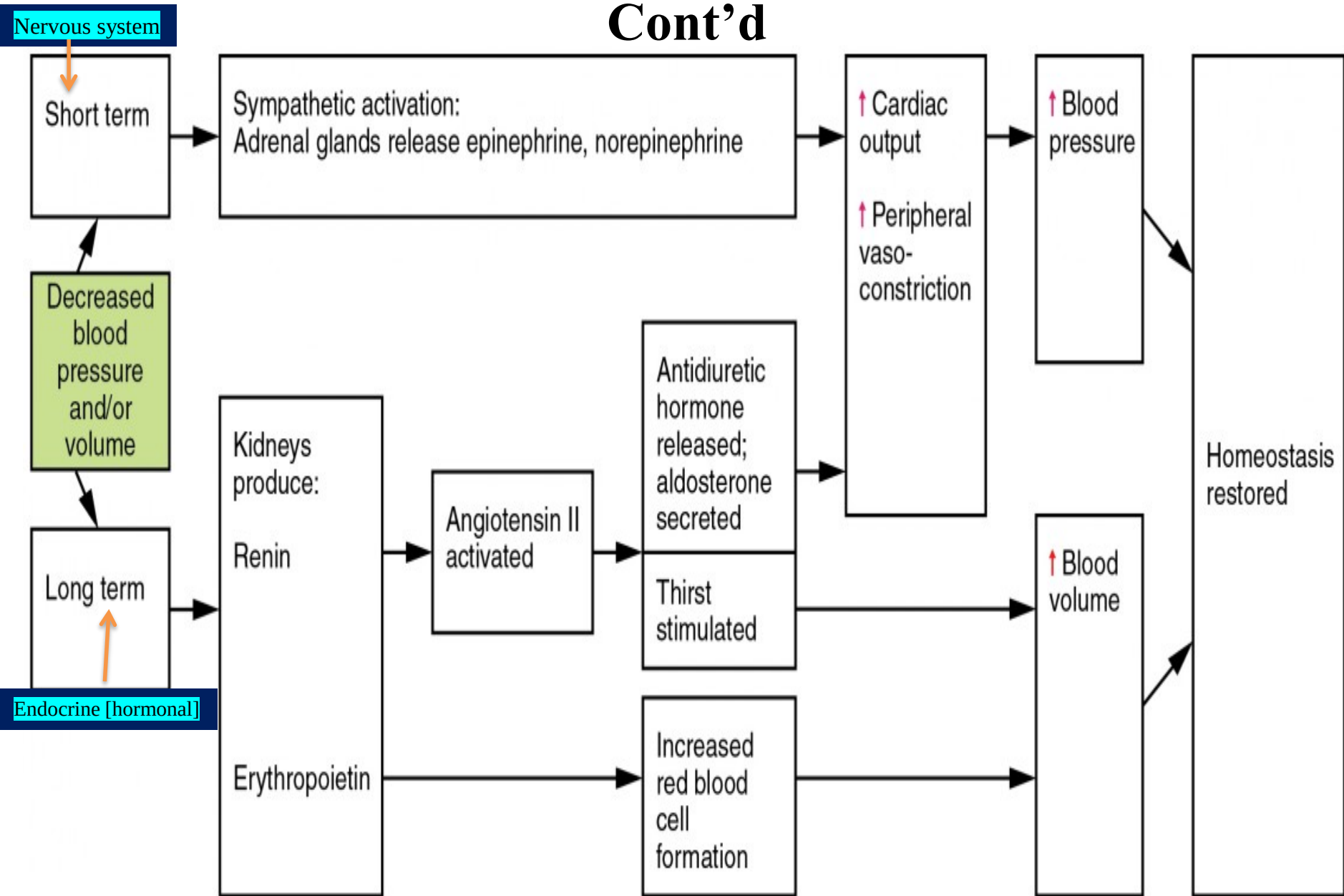
Increased Blood Volume

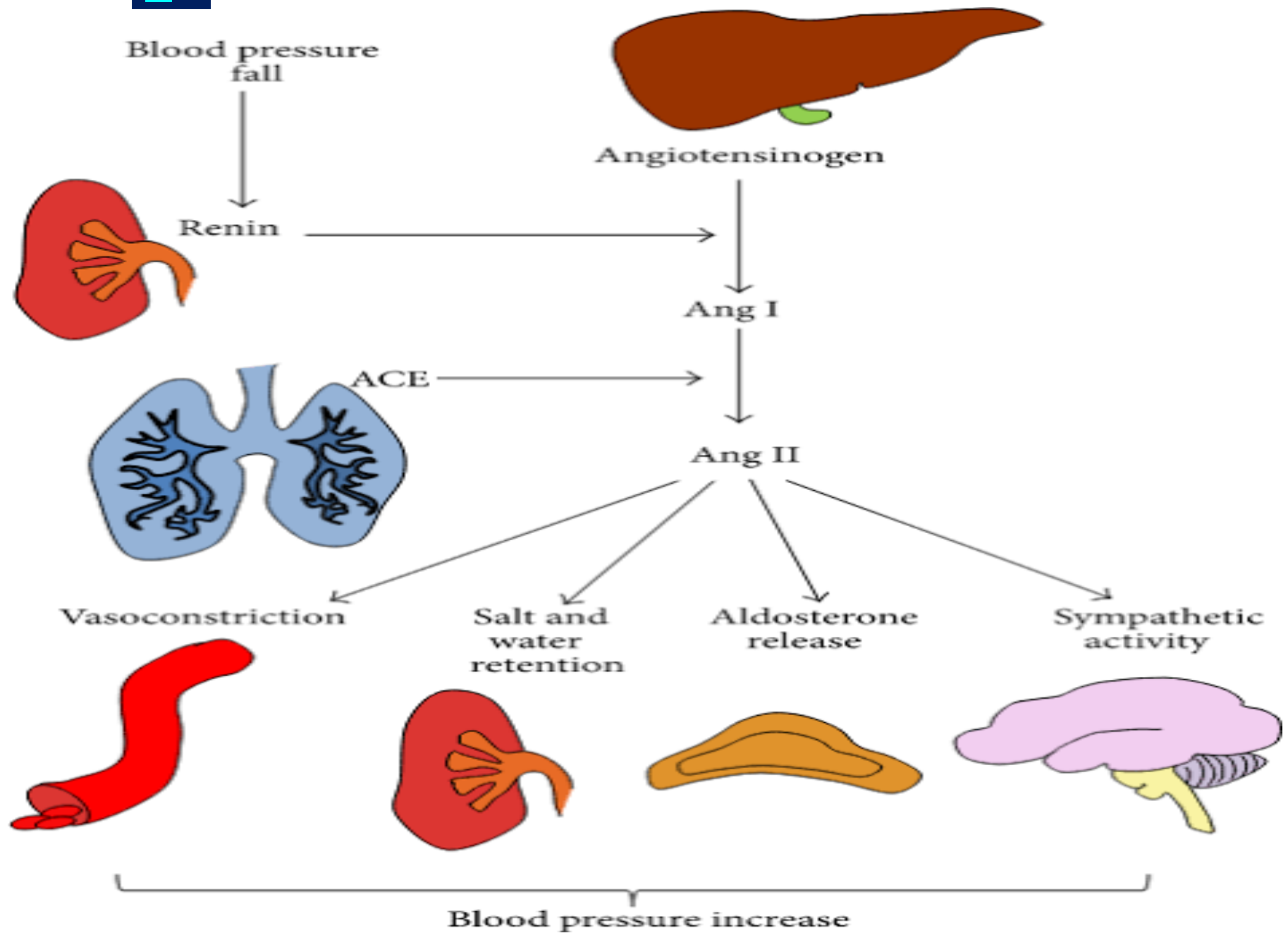
Renal Sodium & Fluid Retention

Sympathetic NS

By Negative Feed Back

Cont'd





Treatment

A. Non-pharmacologic treatment approach of [HT]

- ❑ Freedom from side effect compared with drug therapy
 - ✓ Stop smoking
 - ✓ Lose weight
 - ✓ Decrease in excessive alcohol consumption
 - ✓ Exercise
 - ✓ Reduce salt intake & fatty food

B. Pharmacological treatment of hypertension [HT]

1. Diuretics: deplete salt and blood volume

✓ **Frusemide** , **Spironolactone** , Ethacrinic acid, Triamterene, Amiloride, Mannitol, glycerol, Hydrochlorothiazide, Chlorthalidone, Bendrofluazide

2. Sympathoplegics [sympatholytics]: reduce SNS tone

✓ **Propranolol** , Labetalol, Atenolol , Metoprolol, Guanethidine, Reserpine, Prazosin, Terazosin

3. Angiotensin antagonists [Ang receptor antagonists]: block action of angiotensin

✓ **Losartan** , **Candesartan** , **Eprosartan** , **Valsartan** , **Irbesartan**

4. Vasodilators: dilate resistance and/or capacitance vessels

✓ **Hydralazine** , Sodium nitroprusside , Minoxidil

5. ACE-inhibitors

✓ **Captopril** , **Enalapril** , **Benazepril** , **Ramipril** , **Lisinopril**

6. Renin-inhibitor: inhibit the activity of renin [**Aliskiren**]

The End.